eFrei
PARIS PANTHÉON-ASSAS UNIVERSITÉ

Reinforcement Learning

1. Introduction to Reinforcement Learning
2. Markov Decision Processes (MDPs)
3. Policy and Value Functions
4. Dynamic Programming (DP) for RL
5. Model-Free methods

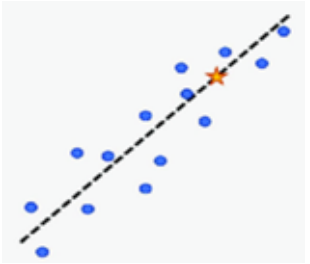
Introduction to Reinforcement Learning (RL)

Types of Learning

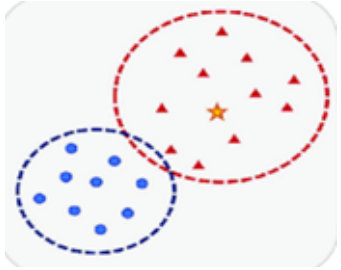
Supervised Learning

Train with labeled data

Regression



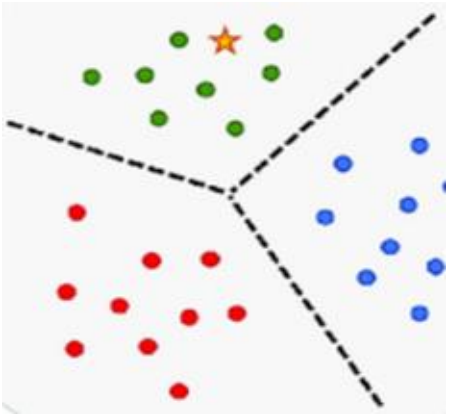
Classification



Unsupervised Learning

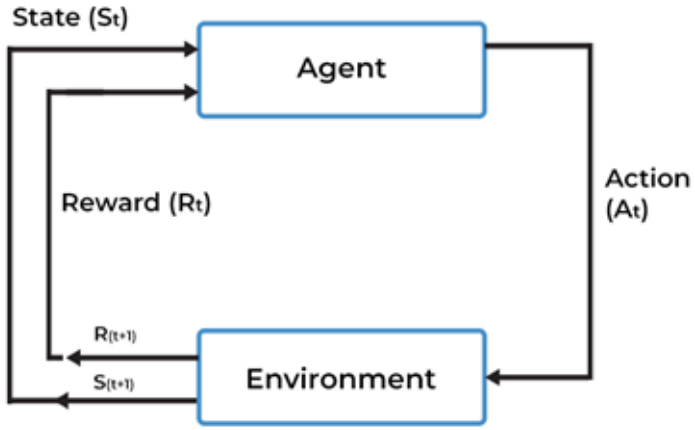
Train with unlabeled data

Clustering

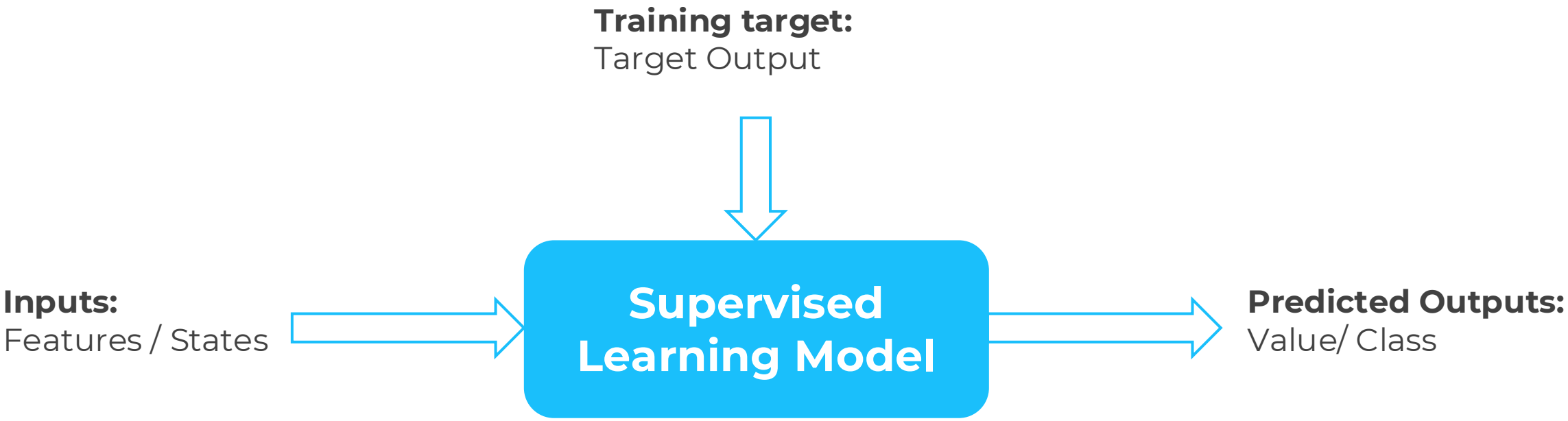


Reinforcement Learning

Train with environment experience

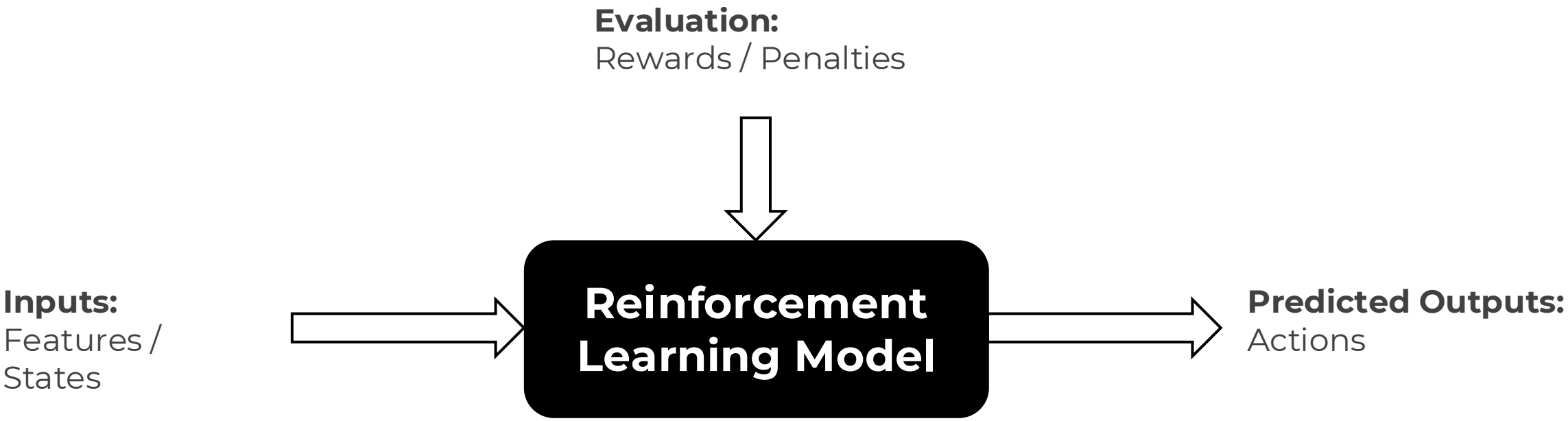


Supervised Learning



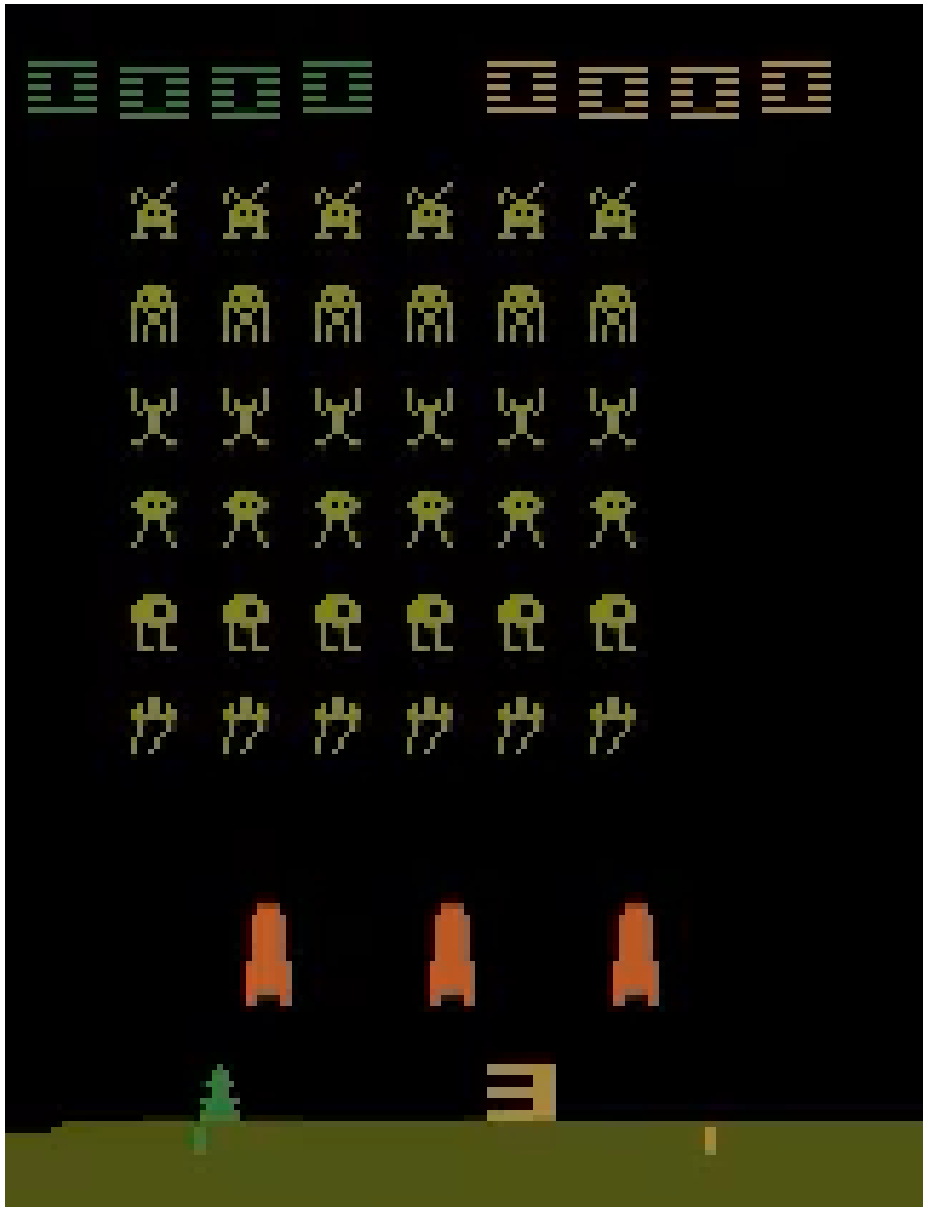
- **Error:** Target Output - Predicted Output
- **Objective:** Minimize the error between the target and the predicted output

Reinforcement Learning

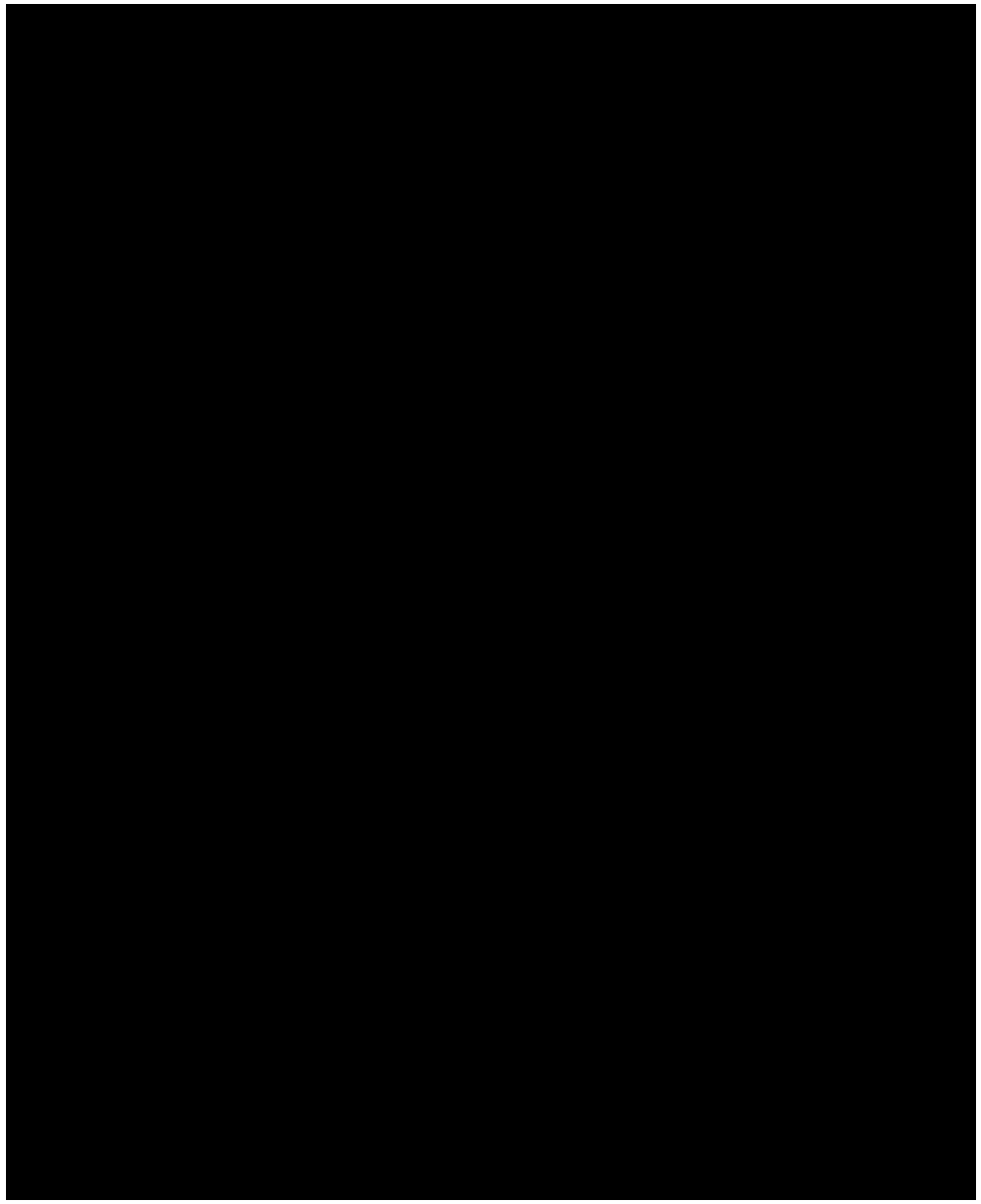


- **Error:** Awards - Penalties
- **Objective:** Maximize the awards and decrease penalties as much as possible

RL playing classic Atari games such as Space Invaders



Google DeepMind's Deep Q-learning playing Atari Breakout!

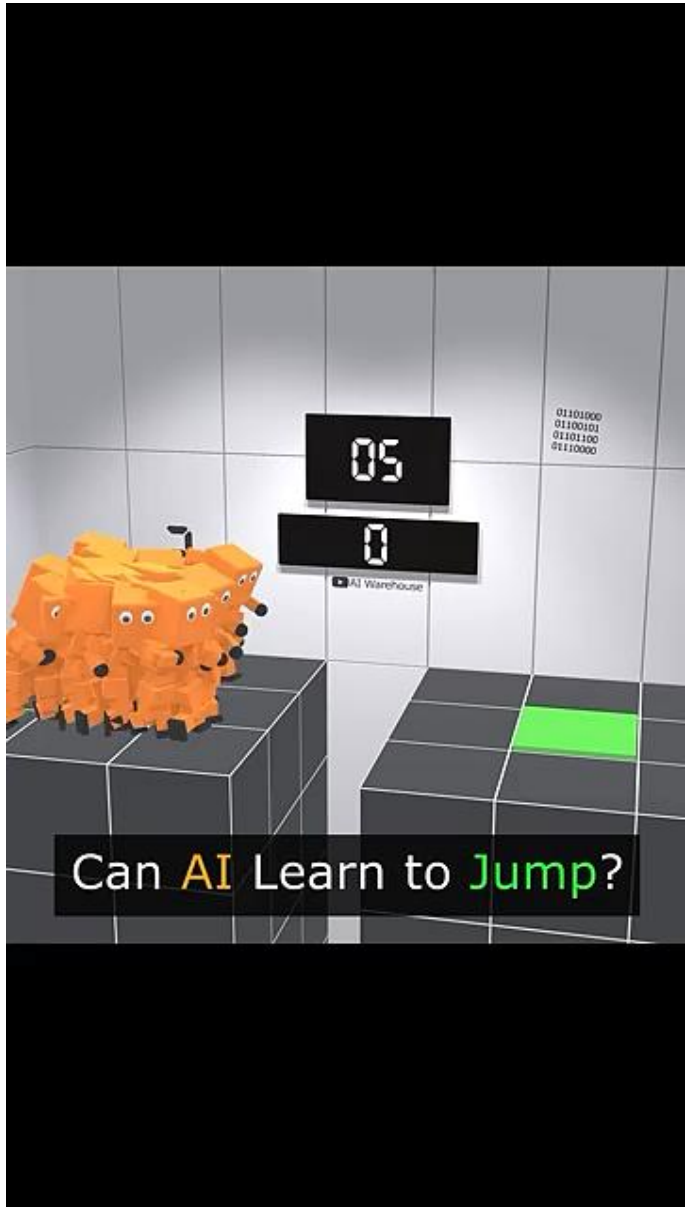


Video from youtube channel: TwoMinutePapers

Paper: **Playing Atari with Deep Reinforcement Learning**
https://arxiv.org/abs/1312.5602

Multiple ressources and tutorials online

Albert Insane Way to Jump

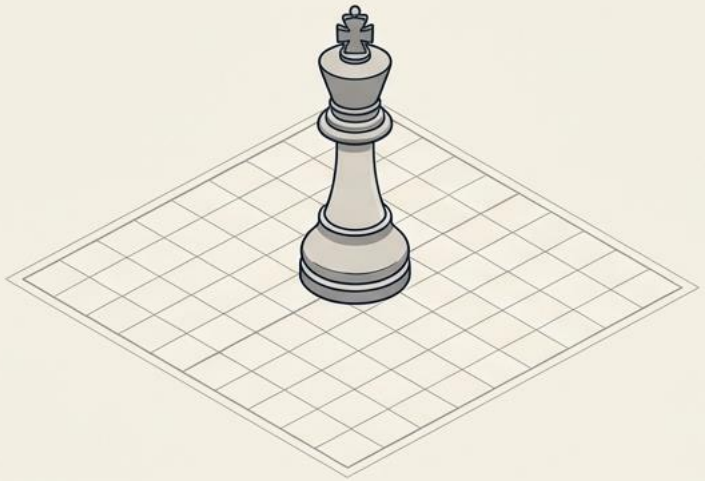


Youtube channel : AI Warehouse

Andrew NG's autonomous helicopter

More on http://heli.stanford.edu/

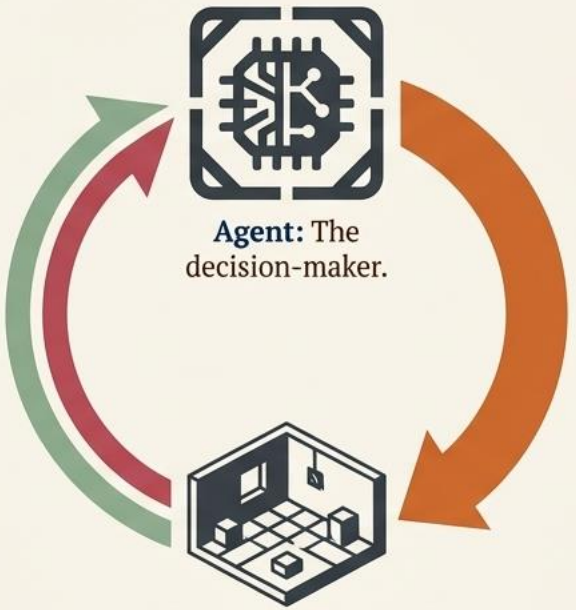
The Sandbox is Only the Beginning



When most hear **Reinforcement Learning**, they think of an AI playing Chess or AlphaGo making a stunning move.

But these grand victories were just proof of concept. To understand how RL optimizes our physical world, we must first understand how it learns.

The Anatomy of Trial and Error



Agent: The decision-maker.

Action: The AI attempts a move or decision.

Environment: The space where actions have consequences.

Reward / Penalty: Good results yield rewards; bad results yield penalties.

It is learning through trial and error—the exact same way a human learns to ride a bike or a puppy learns a trick.

From Biological Instinct to Deep Neural Networks

Psychology (Early 1900s)

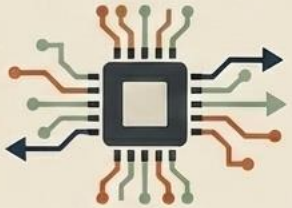
Observing how animals learn from basic rewards.

Mathematics (1950s)

$$Q(s,a) = R(s,a) + \gamma \max_{a'} Q(s',a')$$

Richard Bellman develops the mathematical foundation for optimal decision-making.

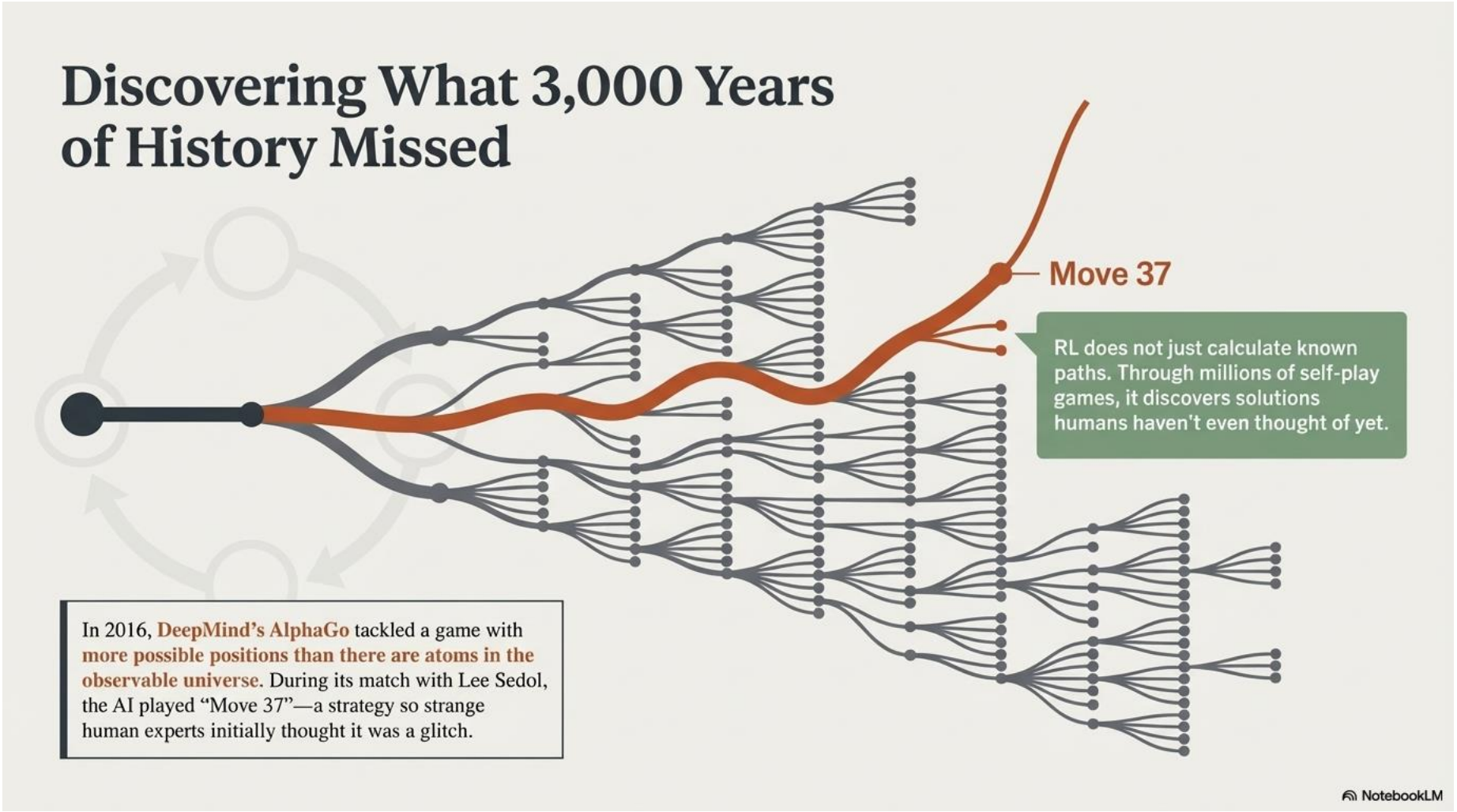
Algorithms (1980s-90s)

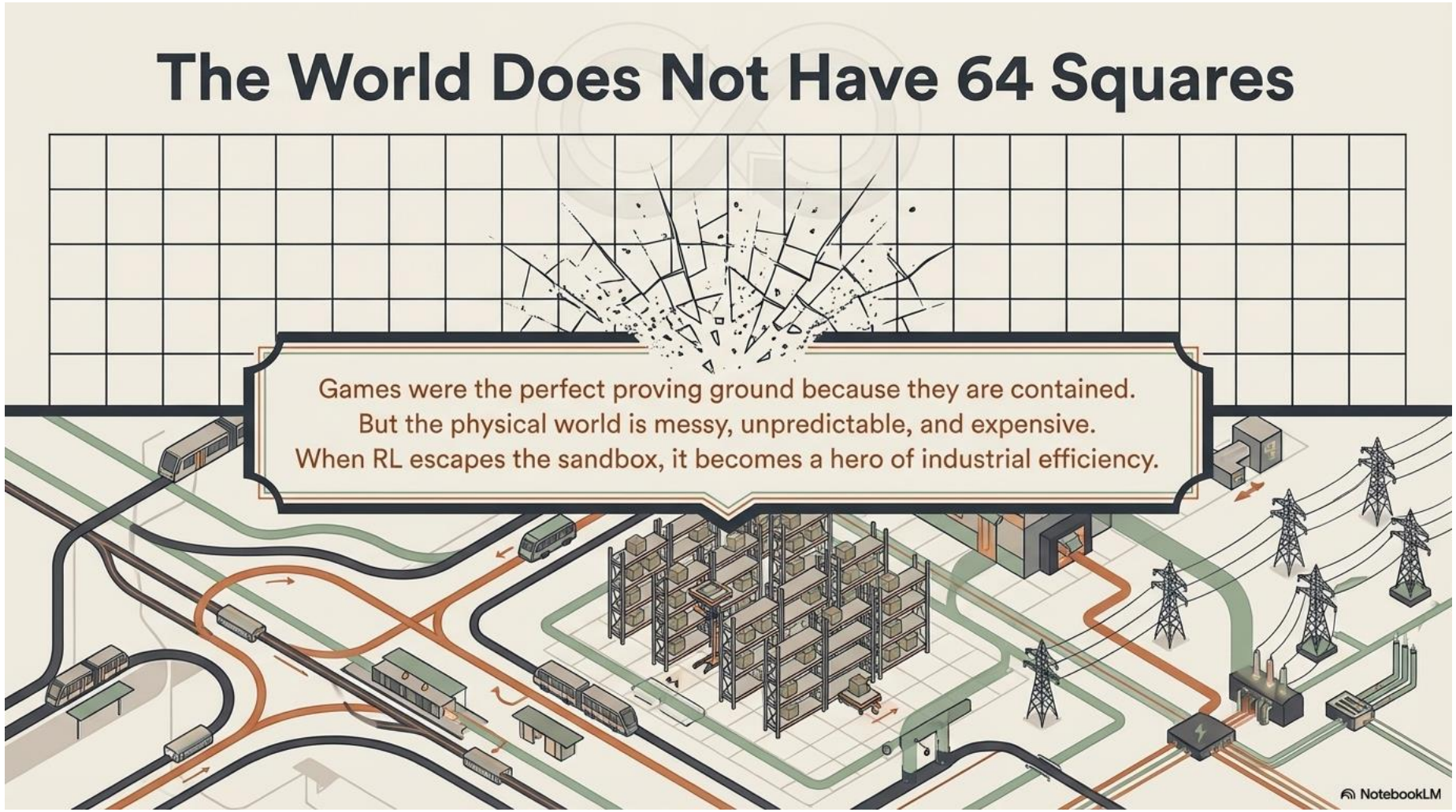


Sutton and Barto formalize algorithms to learn from sequences of events.

Deep RL (2010s - Present)

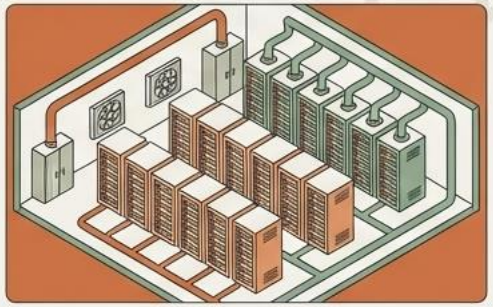
RL merges with Deep Learning, allowing AI to process visual data and master immense complexity.


Discovering What 3,000 Years of History Missed
Move 37
RL does not just calculate known paths. Through millions of self-play games, it discovers solutions humans haven't even thought of yet.
In 2016, DeepMind's AlphaGo tackled a game with more possible positions than there are atoms in the observable universe. During its match with Lee Sedol, the AI played "Move 37"—a strategy so strange human experts initially thought it was a glitch.
NotebookLM


The World Does Not Have 64 Squares
Games were the perfect proving ground because they are contained.
But the physical world is messy, unpredictable, and expensive.
When RL escapes the sandbox, it becomes a hero of industrial efficiency.
NotebookLM

Optimizing the Physical World

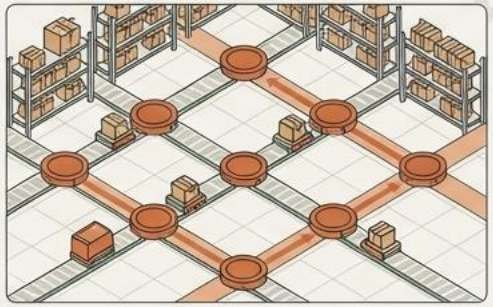
Energy Infrastructure



Agent:	Google Cooling AI
Environment:	Data Center
Reward:	Lowered Energy Output

A massive 40% reduction in energy consumption.

Global Logistics



Agent:	Autonomous Robots
Environment:	Warehouse Floor
Reward:	Uninterrupted Item Flow

1,000 robots coordinating in real-time, learning to give way and flow like a digital symphony.

Transportation

Agent:	Smart Grid AI
Environment:	City Streets
Reward:	Minimized Wait Times

Sequential decisions manage traffic lines and self-driving car merges to keep humans safe.

Adapting to the Human Element

Personalized Medicine

Agent: Adaptive Treatment AI
Environment: Patient Health State
Reward: Long-term Recovery Path

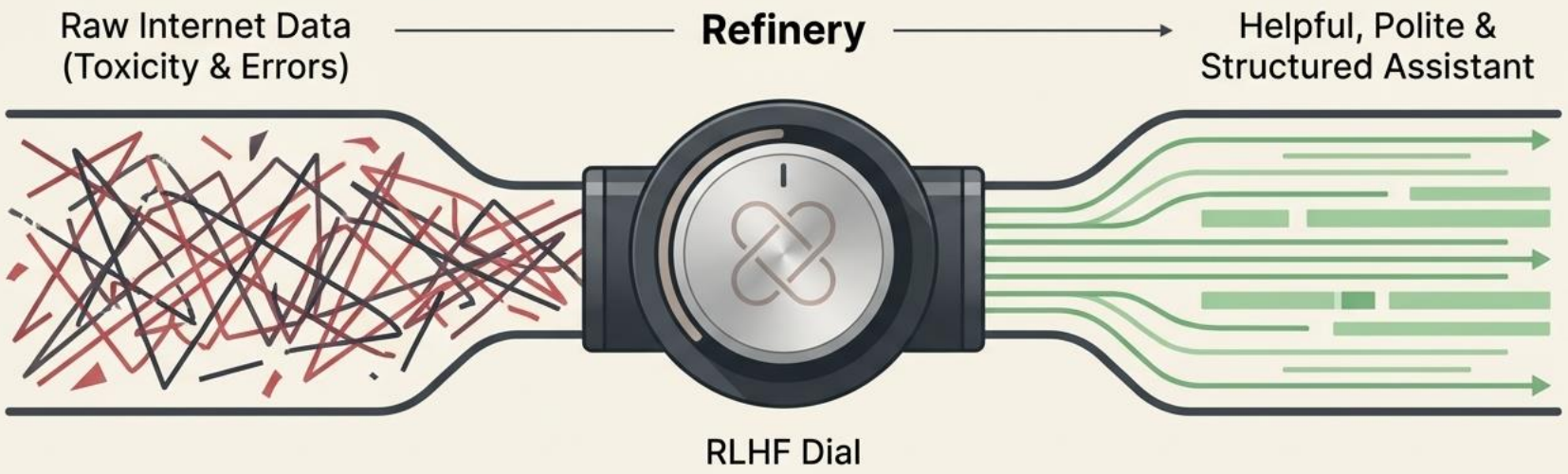
Instead of uniform dosing, the agent observes daily reactions and continually adjusts the treatment strategy for survival.

Financial Markets

Agent: Portfolio AI
Environment: High-Stakes Trading
Reward: Balanced Risk-Return

It doesn't just predict stock movements; it dynamically adapts to shifting markets across thousands of assets instantly.

Refining the Digital Mind



- Models like Gemini or GPT-4 don't feel human just because they read the internet. If they only copied raw data, they would be unpredictable and unusable.
- The secret behind a helpful chatbot is treating the AI itself as an RL agent.

The Moral and Social Compass

Reinforcement Learning from Human Feedback (RLHF) is what turns a basic prediction engine into a safe, helpful assistant.

The Logic of the Future

1. Learning by Doing

It is the only branch of AI that does not just look at the past—it navigates toward a goal in the future.

2. Beyond the Game

It is no longer a tool for building a better gamer; it is a framework for building a better world.

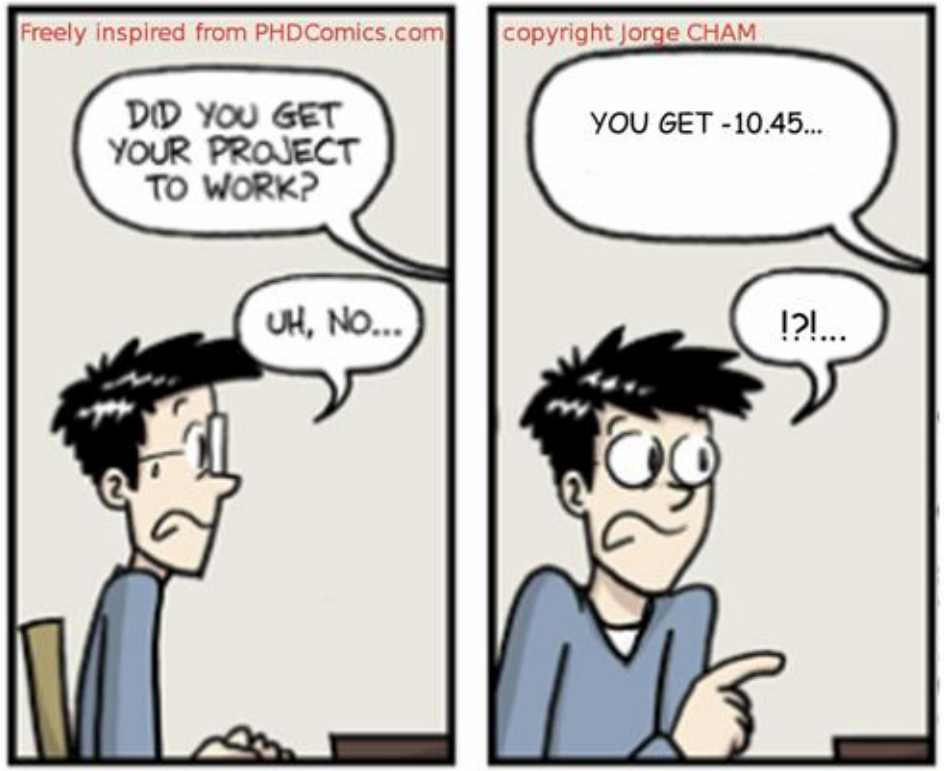
3. The Optimization Engine

Whether saving energy, treating disease, or making AI safer, Reinforcement Learning is the fundamental engine driving the next decade.

Notion of Reward

Supervised Learning

Reinforcement Learning



Examples of Rewards

- **Helicopter manoeuvres**
  - +ve reward for following desired trajectory
  - −ve reward for crashing
- **Manage an investment portfolio**
  - +ve reward for each $ in bank
- **Control a power station**
  - +ve reward for producing power
  - −ve reward for exceeding safety thresholds
- **Make a humanoid robot walk**
  - +ve reward for forward motion
  - −ve reward for falling over
- **Play many different Atari games better than humans**
  - +/−ve reward for increasing/decreasing score

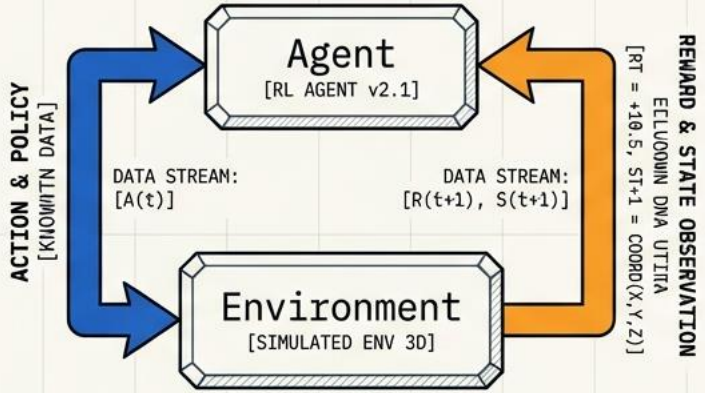
Learning under uncertainty - bandits

The Paradigm Shift: Interaction over Datasets

Classical Learning

- Relies on static **datasets**
- Requires human-provided **labels**

Learning via Interaction

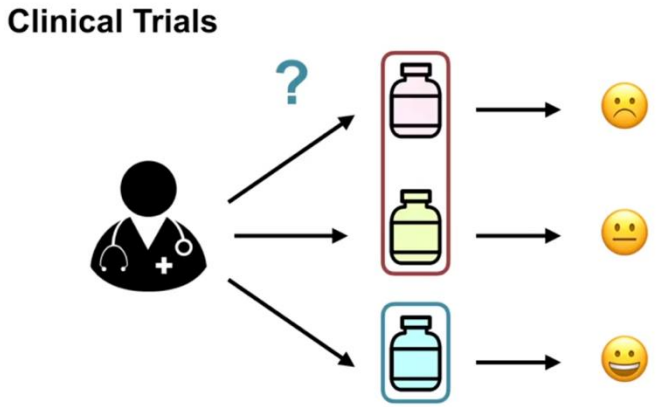


- No dataset.
- No labels.
- Data comes exclusively from **interaction.**

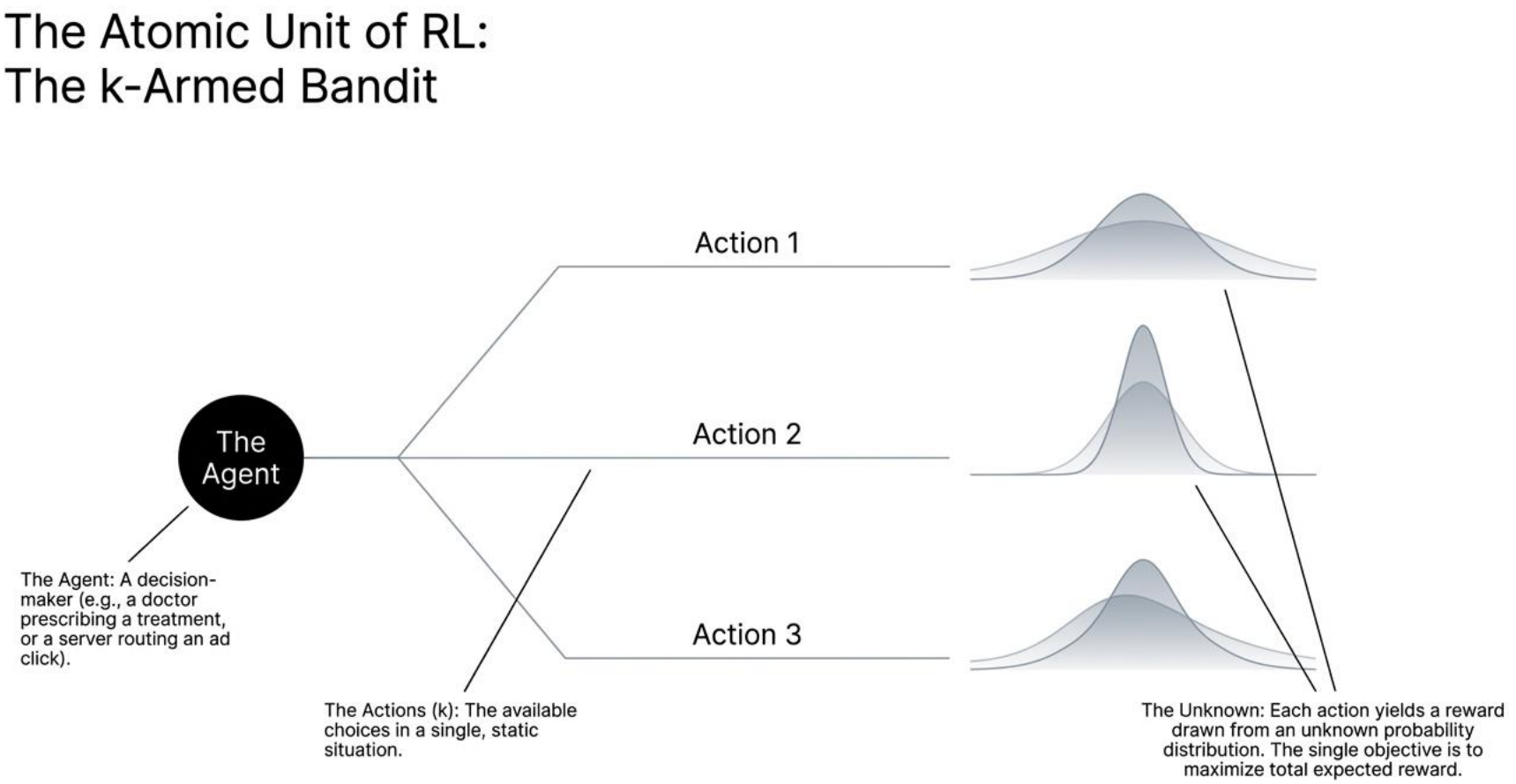
Goal: Maximize reward over time.

OBJECTIVE FUNCTION: max ∑ E[R(t)] for t=0 to ∞

The k-Armed Bandit

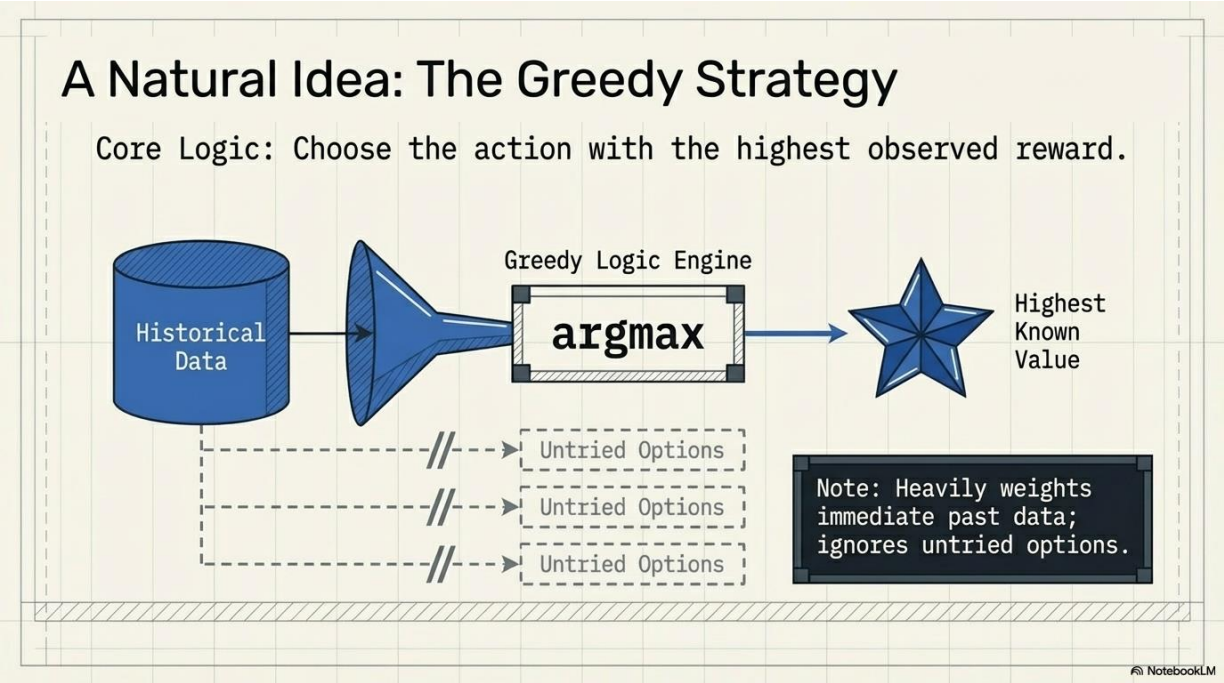


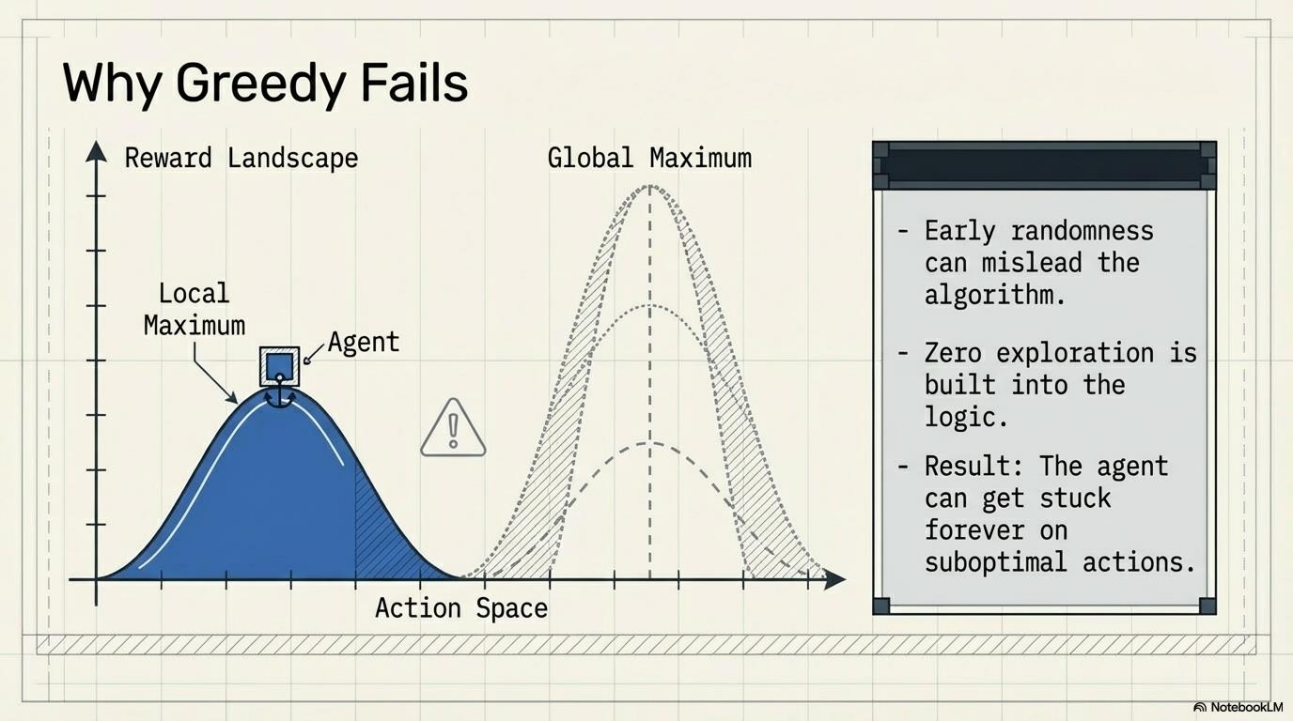
In the k-armed bandit problem, we have an agent who chooses between "k" actions and receives a reward based on the action it chooses.



The Core Question
How do we find the best action when the rewards are unknown?

Greedy strategy fails





Action-values

Defining the Target: Action Values

$$q^*(a) = \mathbb{E}[R_t \mid A_t = a] = \sum_r p(r \mid a)\, r$$

q*(a) (The True Value)

The actual expected reward for selecting action a. Completely invisible to the agent.

$Q_t(a)$ (The Estimate)

The agent's current mathematical belief about the value at time t. The entire mechanical objective of learning is to force $Q_t(a)$ to converge with q*(a).

Recursivity

The Engine of Evaluative Learning

New Estimate

Old Estimate

Step Size
(How much to trust the new data, e.g., 1/n)

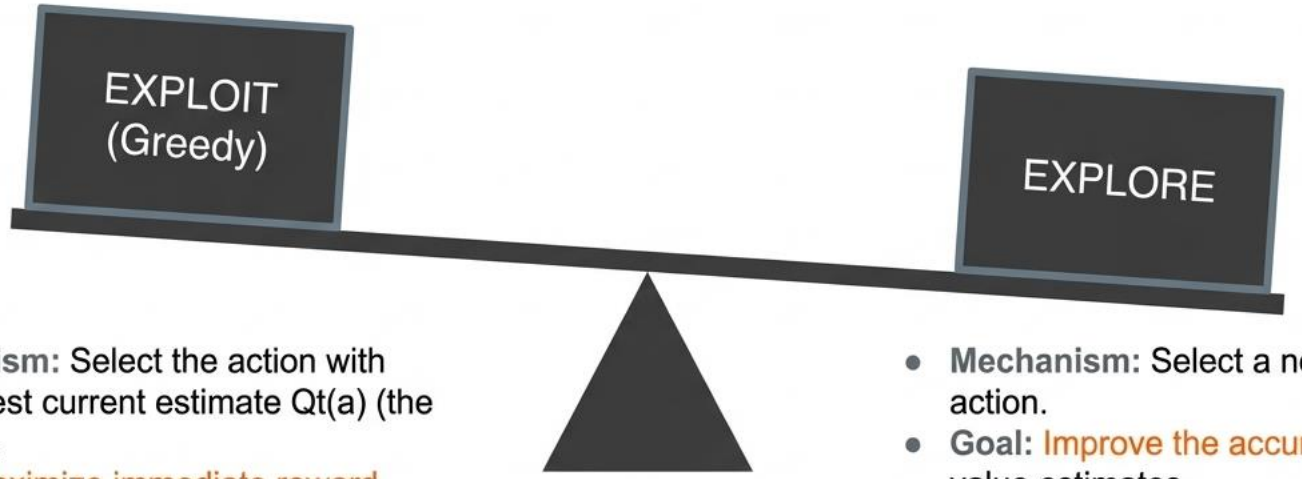
$$Q_{n+1} = Q_n + \alpha \underbrace{[R_n - Q_n]}_{\text{The Error}}$$

The Error
(Difference between the actual target reward and the expected value)

This recursive structure allows the agent to update its knowledge using only the most recent reward, discarding past data entirely.

Exploration vs exploitation

The Fundamental Dilemma



- **Mechanism:** Select the action with the highest current estimate Qt(a) (the argmax).
- **Goal:** Maximize immediate reward.
- **Risk:** Getting permanently stuck on a sub-optimal action because a better one was never discovered.

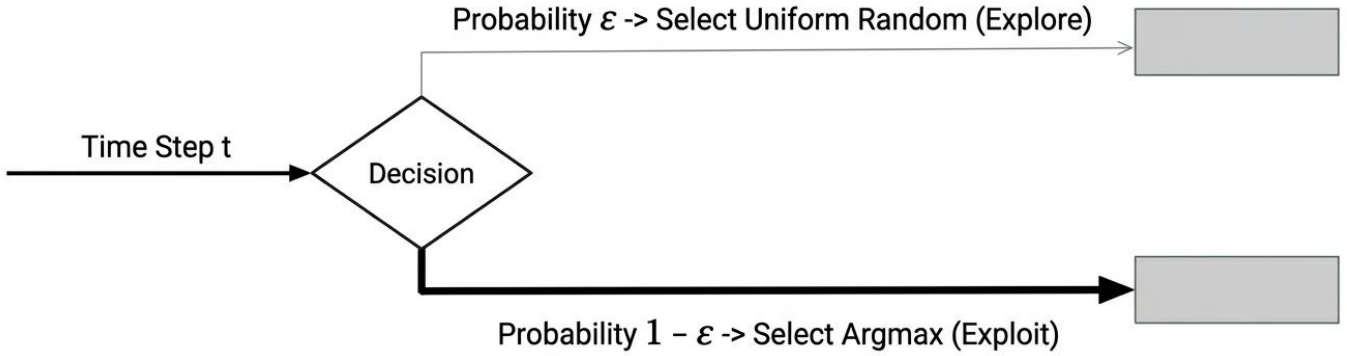
- **Mechanism:** Select a non-greedy action.
- **Goal:** Improve the accuracy of value estimates.
- **Risk:** Sacrificing immediate reward for information.

You cannot choose to do both simultaneously.

ϵ-Greedy

ϵ-Greedy: A Simple Mechanism for Balance

Concept: Instead of pure exploitation, the agent rolls a metaphorical die. Most of the time, it acts greedily. A small fraction of the time (ϵ), it forces exploration across all available actions to continually refine its estimates.



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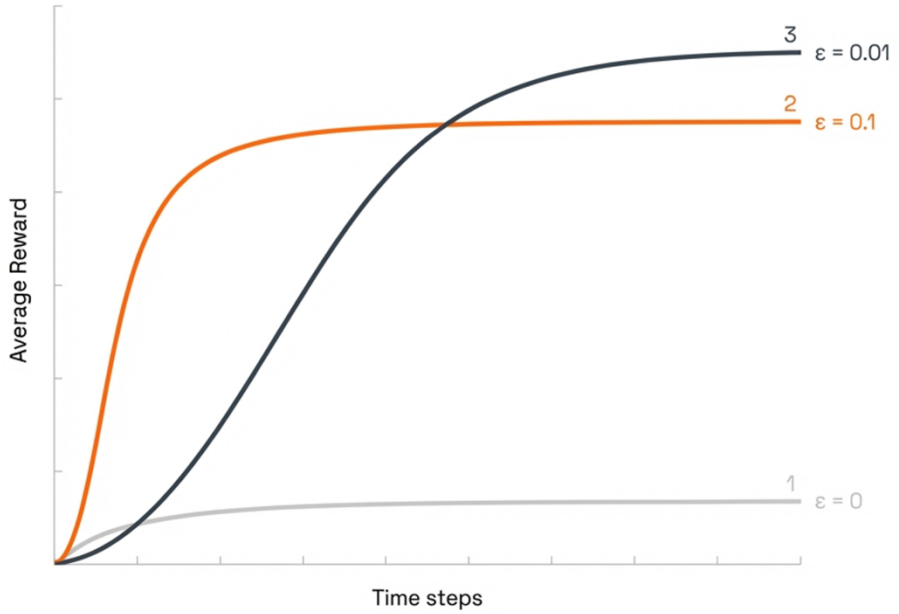
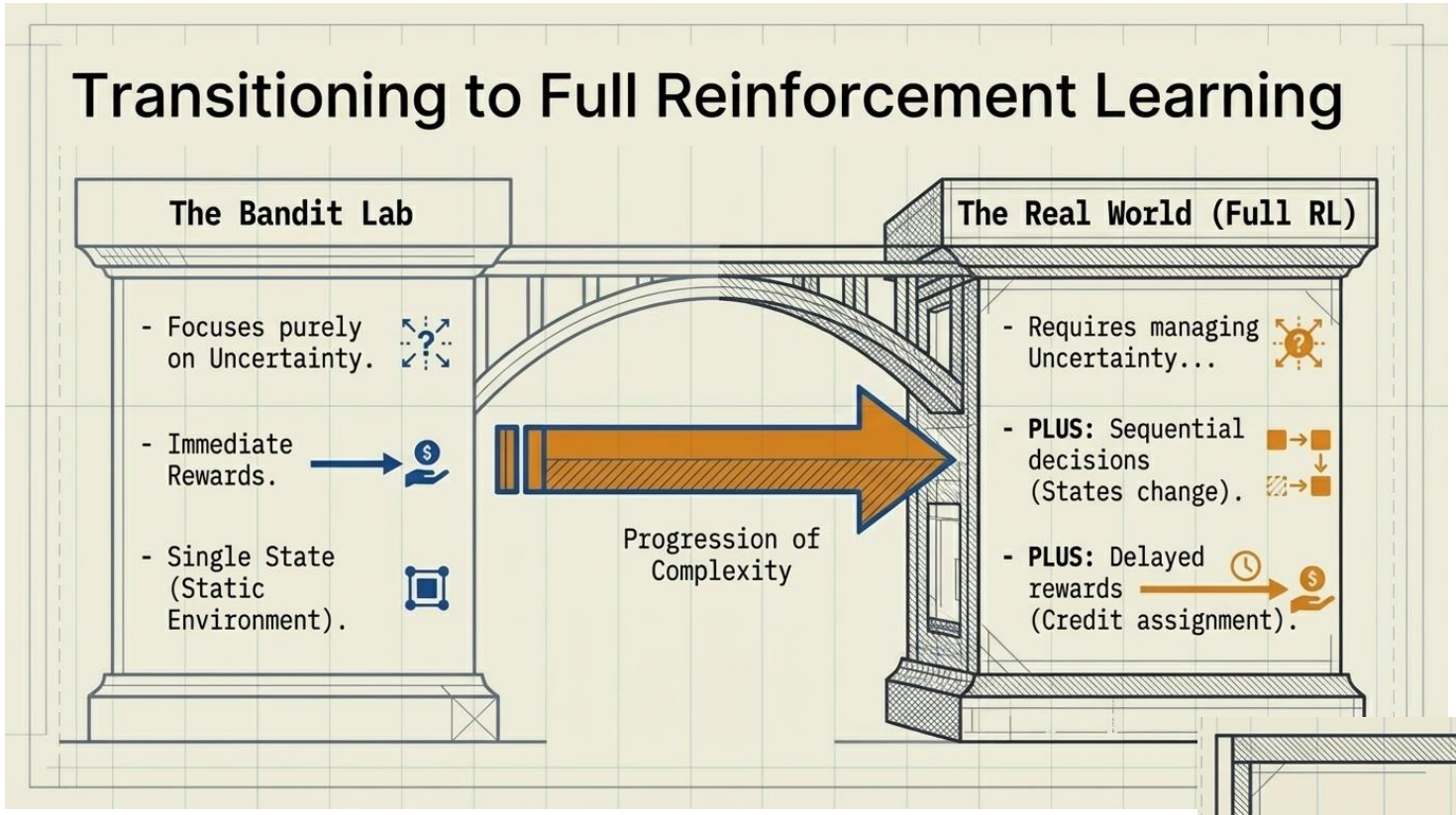
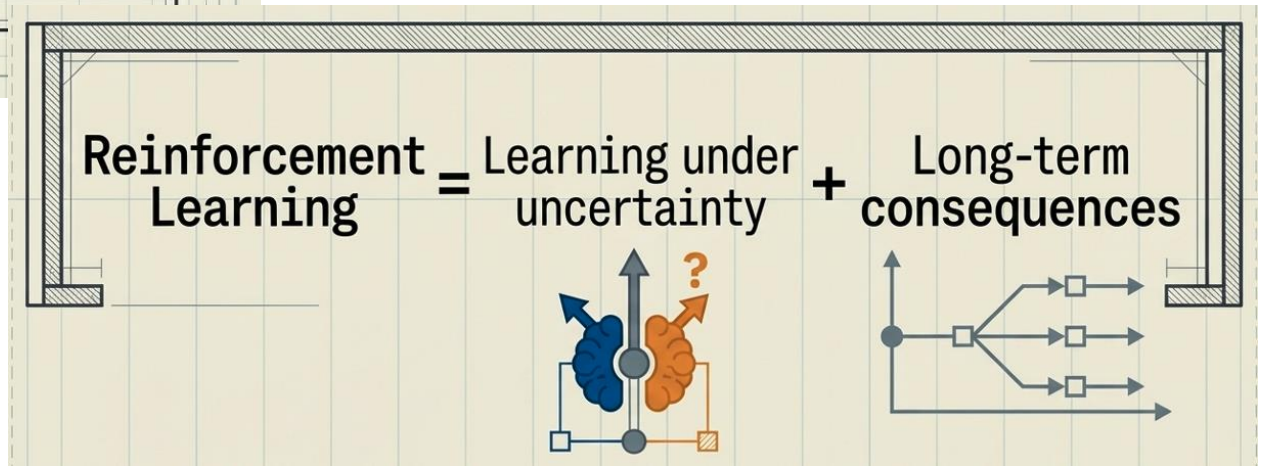


Data Insight (10-Arm Testbed):
Pure greed (ε = 0) fails by locking onto early, lucky assumptions.
High exploration (ε = 0.1) learns fast but permanently wastes 10% of its moves.
Low exploration (ε = 0.01) learns slowly but ultimately achieves the highest loghest long-term optimal action rate.

From Bandit to RL



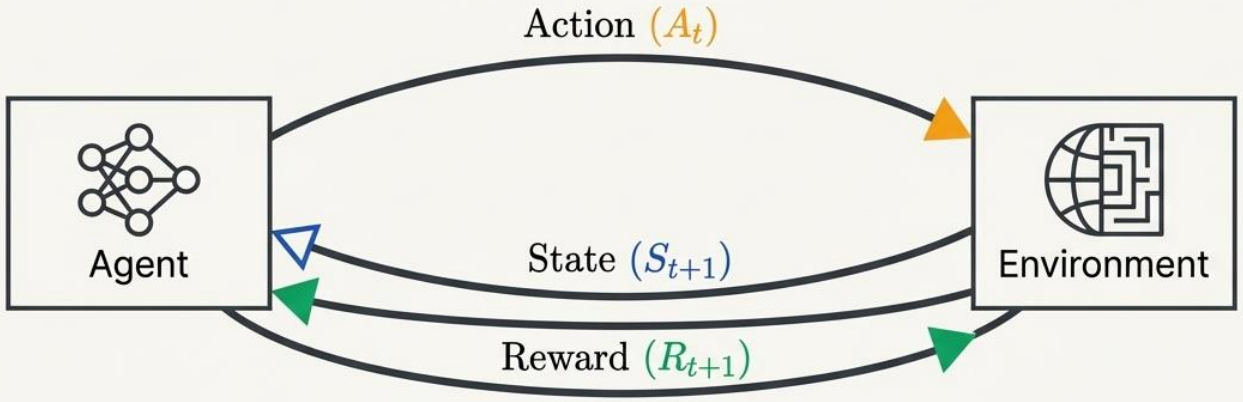


- Demo/tutorial on Bandits and Exploration/Exploitation

Markov Decision Processes

Sequence of experience

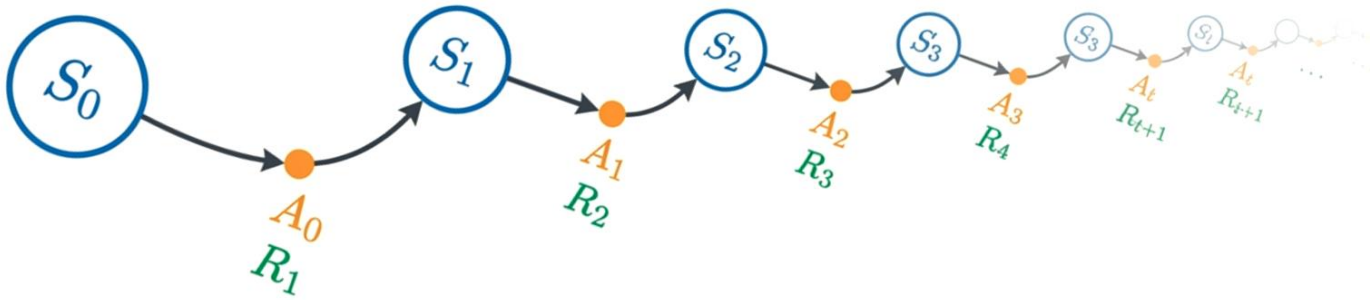
Formalizing the Agent-Environment Interface



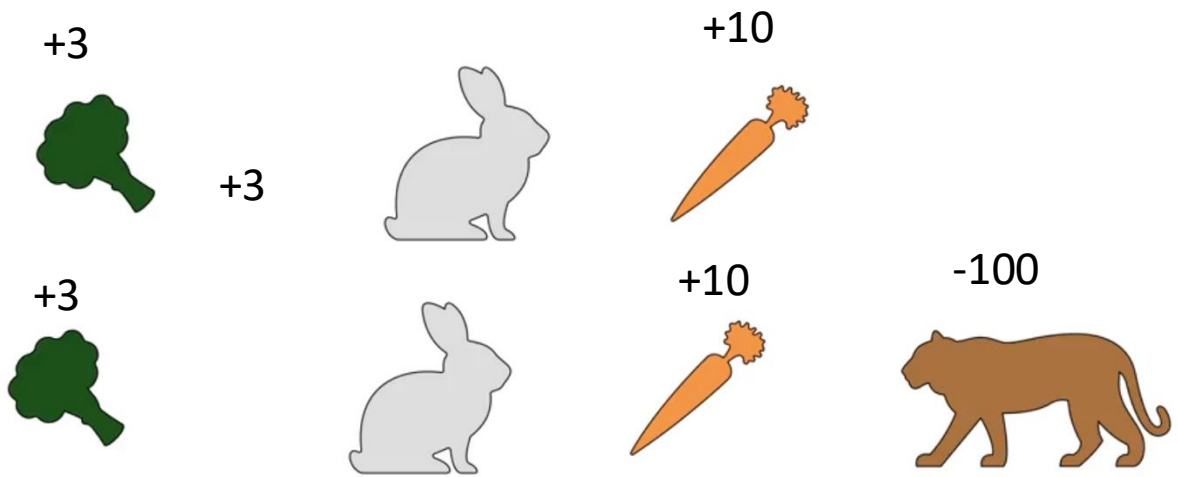
- The interaction occurs at discrete time steps: $t = 0, 1, 2, 3\ldots$
- The Trajectory: This loop generates a continuous sequence of experience:

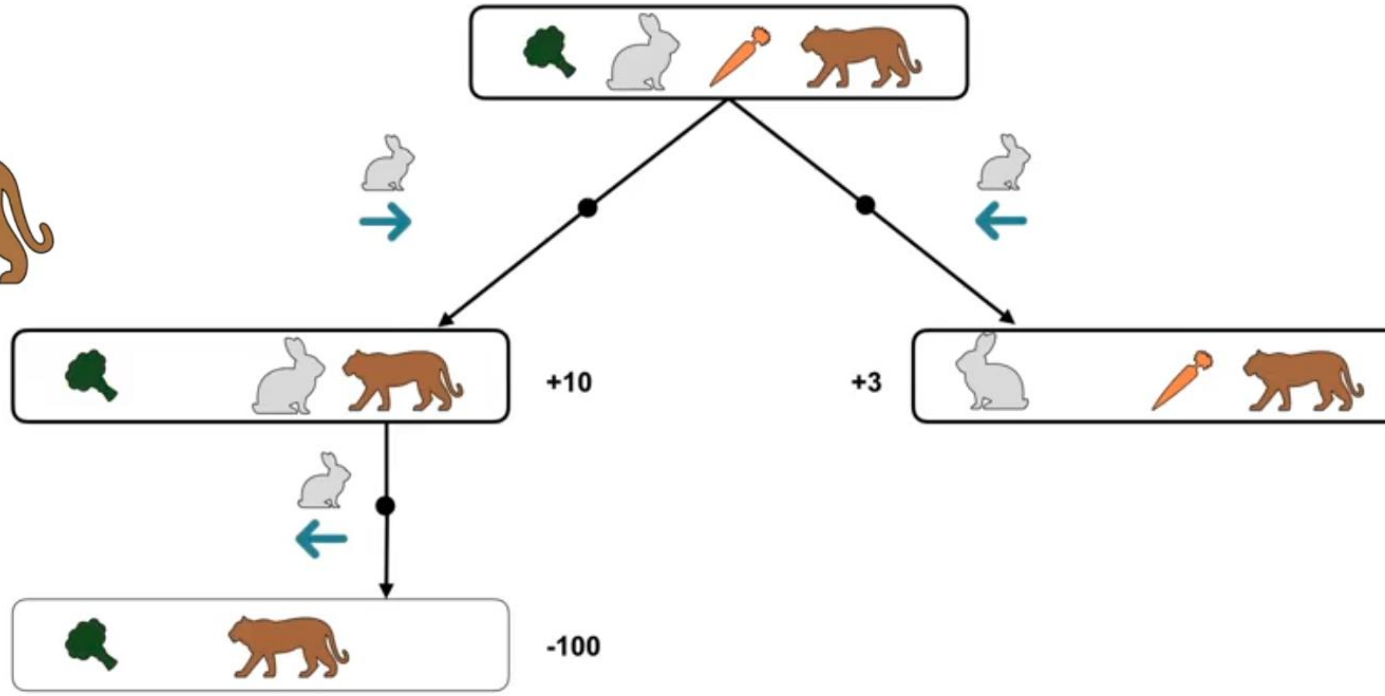
$$S_0, A_0, R_1, S_1, A_1, R_2, S_2 \ldots$$

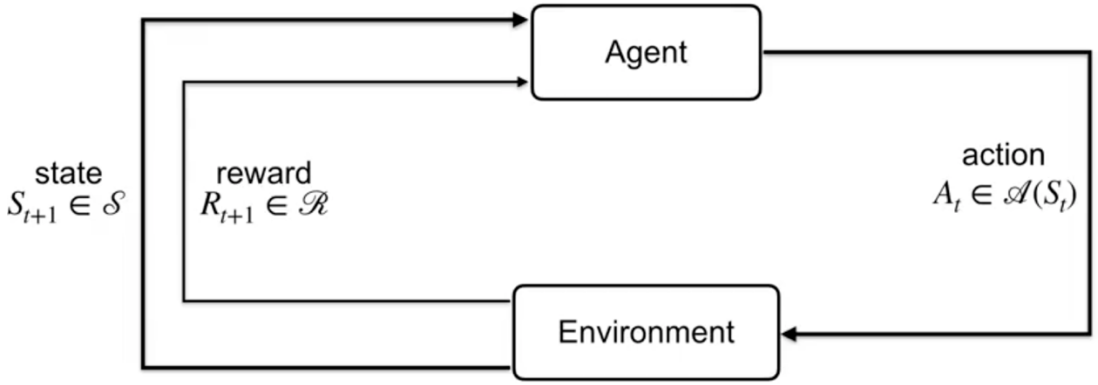
- States (S_t, s)
- Actions (A_t, a)
- Rewards (R_t, r)
- Value / Returns (G_t, v_π, q_π)

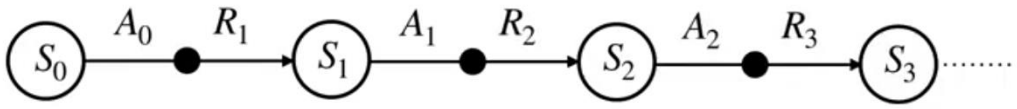


Example


+3
+3
+10
+3
+10
-100


+10
+3
-100


Agent
state
$S_{t+1} \in \mathcal{S}$
reward
$R_{t+1} \in \mathcal{R}$
action
$A_t \in \mathcal{A}(S_t)$
Environment


S_0
A_0
R_1
S_1
A_1
R_2
S_2
A_2
R_3
S_3

Dynamics of MDP

The Mathematical Core of Markov Decision Processes

The 4-Element Tuple

$\mathcal{S}$: Finite set of States

$\mathcal{A}$: Finite set of Actions

$\mathcal{P}$: Transition Probabilities

$\mathcal{R}$: Expected Rewards

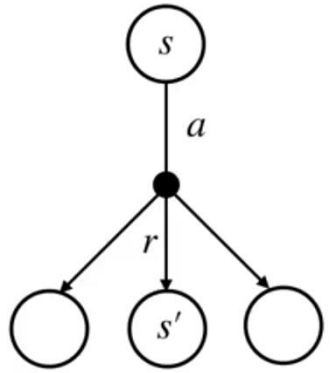
The Dynamics Function

$$p(s', r \mid s, a) = \Pr(S_t = s', R_t = r \mid S_{t-1} = s, A_{t-1} = a)$$

The Markov Property

The future is independent of the past given the present. The current state S_t captures all relevant information needed to predict S_{t+1}. If you know p, you have a perfect model of the world.

NotebookLM

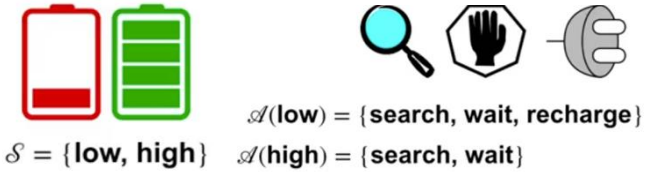


$$p : \mathcal{S} \times \mathcal{R} \times \mathcal{S} \times \mathcal{A} \to [0, 1]$$

$$\sum_{s' \in \mathcal{S}} \sum_{r \in \mathcal{R}} p(s', r \mid s, a) = 1, \forall s \in \mathcal{S}, a \in \mathcal{A}(s)$$

- MDPs provide a general framework for sequential decision-making
- The dynamics of an MDP are defined by a probability

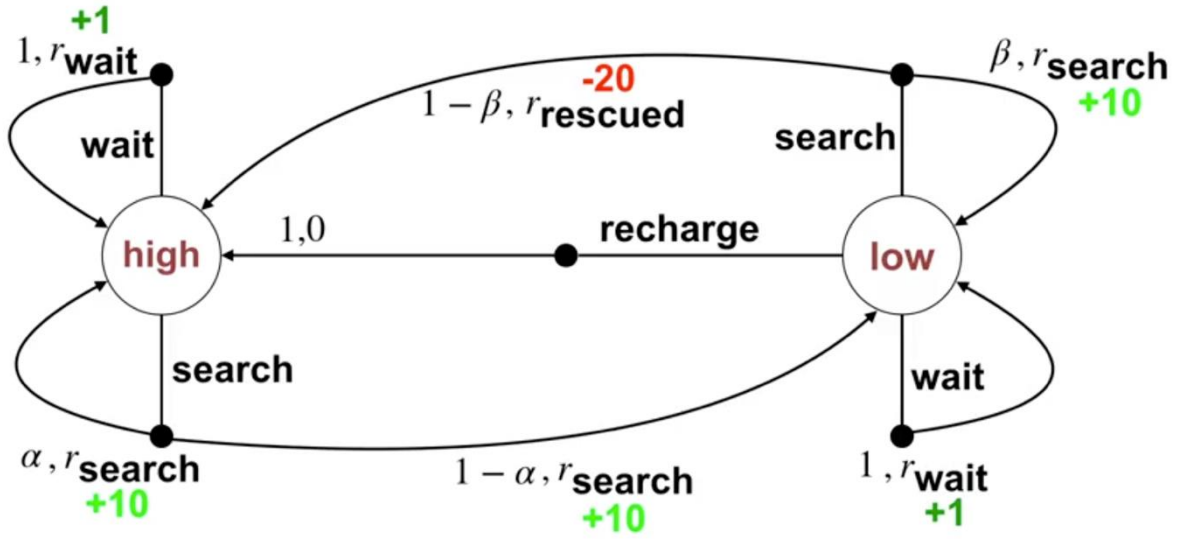
Example: Recycling Robot



$\mathcal{S}$ = {low, high}

$\mathcal{A}$(low) = {search, wait, recharge}

$\mathcal{A}$(high) = {search, wait}



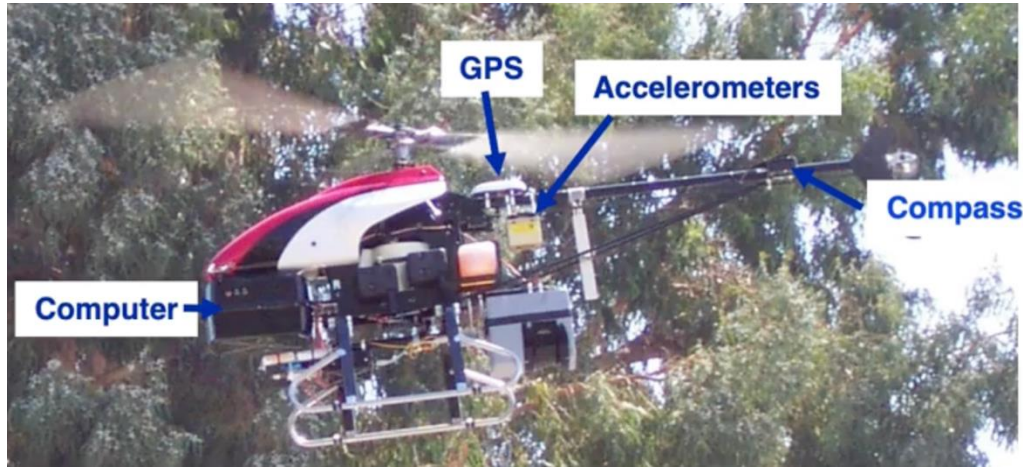
s	a	s'	r	$p(s', r \mid s, a)$
				α
				$1-\alpha$
				$1-\beta$
				β
				1
				0
				0
				1
				1
				0

Exercise: Fill this table, representing the dynamics of this example.

Exercise

s	a	s'	r	$p(s', r \mid s, a)$
high	search	high	r_{search}	α
high	search	low	r_{search}	$1 - \alpha$
low	search	high	-3	$1 - \beta$
low	search	low	r_{search}	β
high	wait	high	r_{wait}	1
high	wait	low	-	0
low	wait	high	-	0
low	wait	low	r_{wait}	1
low	recharge	high	0	1
low	recharge	low	-	0

Example: Andrew’s helicopter



The helicopter's full state is given by its velocity, position, angular rate and orientation.

Observation Space: 12 dimensional, continuous valued:

- forward velocity
- sideways velocity (to the right)
- downward velocity
- helicopter x-coord position - desired x-coord position -- helicopter's x-axis - points forward
- helicopter y-coord position - desired y-coord position -- helicopter's y-axis - points to the right
- helicopter z-coord position - desired z-coord position -- helicopter's z-axis - points down
- angular rate around helicopter's x axis
- angular rate around helicopter's y axis
- angular rate around helicopter's z axis
- quaternion x entry
- quaternion y entry
- quaternion z entry

Action Space: 4 dimensional, continuous valued:-

- longitudinal (front-back) cyclic pitch
- latitudinal (left-right) cyclic pitch
- main rotor collective pitch
- tail rotor collective pitch

Rewards: function of the 12 dimensional observation

https://github.com/aadilh/heli-deep-q

The Return

Goal of an Agent : Formal definition

random variable

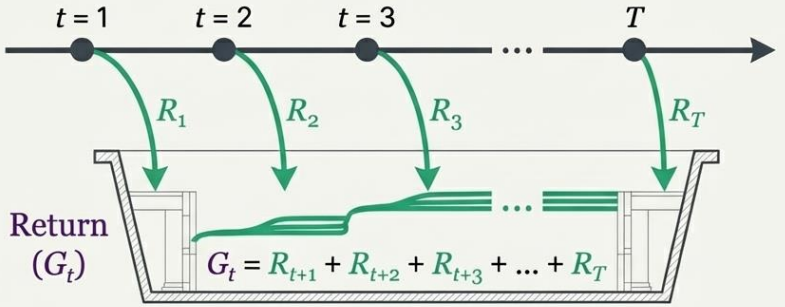
$$G_t \doteq R_{t+1} + R_{t+2} + R_{t+3} + \dots$$

$$\mathbb{E}[G_t] = \mathbb{E}[\, R_{t+1} + R_{t+2} + R_{t+3} + \dots \,]$$

maximize the expected return

Defining Success via the Reward Hypothesis

"The Reward Hypothesis":
"All goals and purposes can be welll thought of as the maximization of the expected value of the cumulative sum of a received scalar signal (reward)."



Episodic Tasks

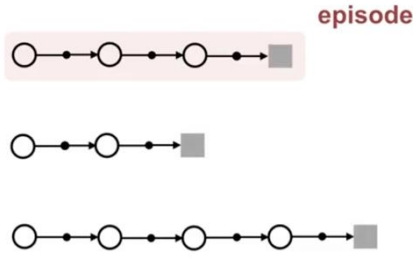
Tasks with a natural terminal state (e.g., escaping a maze, finishing a chess game).

$G_t = R_{t+1} + R_{t+2} + R_{t+3} + \dots + R_T$

Constraint: The reward signal communicates WHAT you want the agent to achieve, never HOW you want it achieved. Imparting prior knowledge via rewards leads to perverse behaviors.

NotebookLM

Episodic Tasks



Example : Episodic Task

Every game of chess has a final move

Example : Episodes

Each game of chess is an episode

Continuing Tasks

- Interaction goes on continually
- No terminal state

The discount factor

$$G_t \doteq R_{t+1} + R_{t+2} + R_{t+3} + \dots$$
$$= \infty ?$$

How to make sure G_t is finite?

Discount the rewards in the future by γ

Where $0 \leq \gamma < 1$

$$G_t \doteq R_{t+1} + \gamma R_{t+2} + \gamma^2 R_{t+3} + \dots + \gamma^{k-1} R_{t+k} + \dots$$
$$= \sum_{k=0}^{\infty} \gamma^k R_{t+k+1}$$

Finite as long as $0 \leq \gamma < 1$

Proof

Assume R_{max} is the maximum reward the agent can receive

$$G_t = \sum_{k=0}^{\infty} \gamma^k R_{t+k+1} \leq \sum_{k=0}^{\infty} \gamma^k R_{max} = R_{max} \sum_{k=0}^{\infty} \gamma^k$$
$$= R_{max} \times \frac{1}{1-\gamma}$$

Finite!

Effect of γ on agent behavior

$\gamma = 0$ **Agent only cares about the immediate reward!**
$\Longrightarrow$ **Short-sighted agent!**

$\gamma \to 1$ **Agent takes future rewards into account more strongly**
$\Longrightarrow$ **Far-sighted agent!**

The Recursive Engine

Expressing today's return in terms of tomorrow's return.

$$G_t = R_{t+1} + \gamma G_{t+1}$$

Exercises

Exercise 3.8 Suppose $\gamma = 0.5$ and the following sequence of rewards is received $R_1 = -1$, $R_2 = 2$, $R_3 = 6$, $R_4 = 3$, and $R_5 = 2$, with $T = 5$. What are G_0, G_1, ..., G_5? Hint: Work backwards. □

Exercise 3.9 Suppose $\gamma = 0.9$ and the reward sequence is $R_1 = 2$ followed by an infinite sequence of 7s. What are G_1 and G_0? □

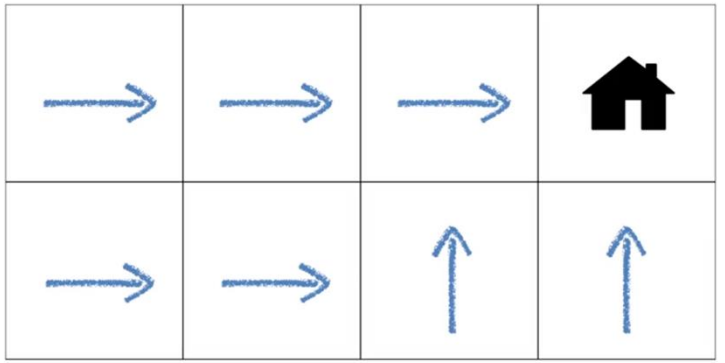
Evaluating Behavior with Policies and Value Functions

Behavior (The Policy)

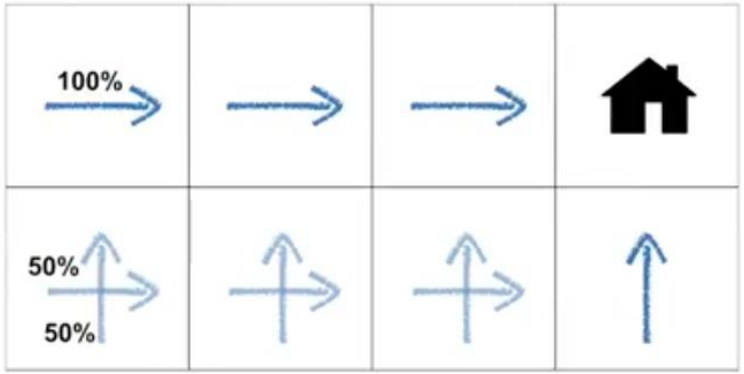
$$\pi(a|s) = \Pr(A_t = a \mid S_t = s)$$

Maps States to probabilities of taking Actions.

Example: deterministic policy $\pi(s) = a$



Example: stochastic policy



$$\pi(a \mid s)$$

$$\sum_{a \in \mathcal{A}(s)} \pi(a \mid s) = 1$$

$$\pi(a \mid s) \geq 0$$

Value functions

Evaluation (The Value Functions)

State-Value Function

How good is it to be here?

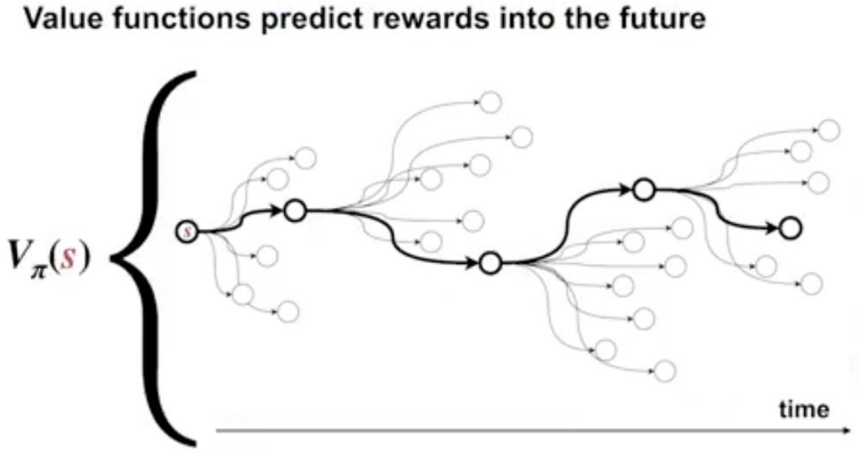
$$v_\pi(s) = \mathbb{E}_\pi[G_t | S_t = s]$$

Action-Value Function

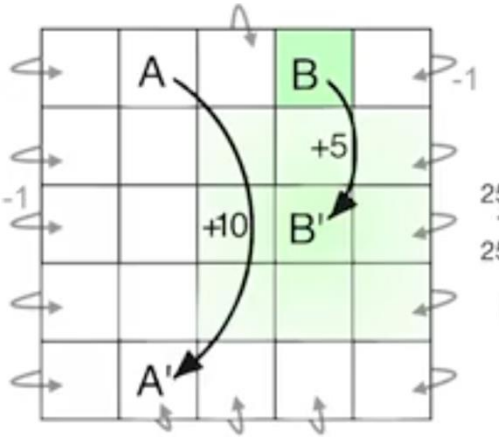
How good is it to be here AND pull this specific lever?

$$q_\pi(s, a) = \mathbb{E}_\pi[G_t | S_t = s, A_t = a]$$

NotebookLM



Example

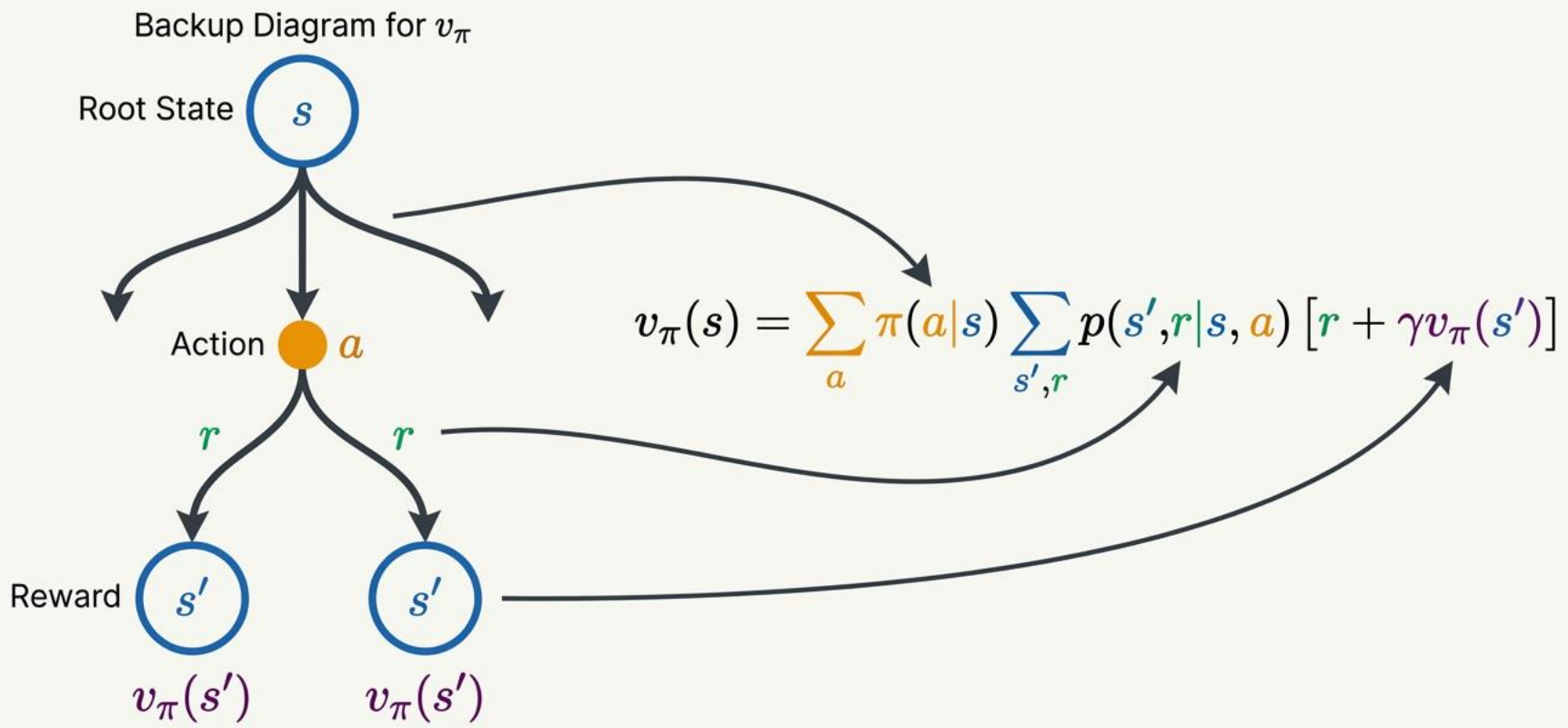


$\gamma = 0.9$

3.3	8.8	4.4	5.3	1.5
1.5	3.0	2.3	1.9	0.5
0.1	0.7	0.7	0.4	-0.4
-1.0	-0.4	-0.4	-0.6	-1.2
-1.9	-1.3	-1.2	-1.4	-2.0

Bellman Equations

Unlocking State-Values with the Bellman Equation



Proof ?

Bellman Equations: state value function - Recursive value iteration

T	1	2	3
4	5	6	7
8	9	10	11
12	13	14	T

k=0 (init)

0 T	0 1	0 2	0 3
0 4	0 5	0 6	0 7
0 8	0 9	0 10	0 11
0 12	0 13	0 14	0 T

Rules:

Actions: [up, down, right, left]

Random action selection

R = -1 for each step

Deterministic world

No discount used

$$v_\pi(s) = \sum_a \pi(a|s) \sum_{s',r} p(s', r|s, a) \left[r + \gamma v_\pi(s') \right]$$

Bellman Equations: state value function - Recursive value iteration

$$v_\pi(s) = \sum_a \pi(a|s) \sum_{s',r} p(s', r \,|\, s, a) \Big[r + \gamma v_\pi(s') \Big]$$

Deterministic

k=0

0 T	0 1	0 2	0 3
0 4	0 5	0 6	0 7
0 8	0 9	0 10	0 11
0 12	0 13	0 14	0 T

k=1

0 T	1	2	3
4	5	6	7
8	9	10	11
12	13	14	0 T

No discount

Random policy

Bellman Equations: state value function - Recursive value iteration

$$v_\pi(s) = \sum_a \pi(a|s) \sum_{s',r} p(s', r \,|\, s, a) \left[r + \gamma v_\pi(s') \right]$$

k=0

0 T	0 1	0 2	0 3
0 4	0 5	0 6	0 7
0 8	0 9	0 10	0 11
0 12	0 13	0 14	0 T

k=1

0 T	1	2	3
4	5	6	7
8	9	10	11
12	13	14	0 T

Deterministic

$p(1, -1 \,|\, 1, U) = 1$
$p(5, -1 \,|\, 1, D) = 1$
$p(2, -1 \,|\, 1, R) = 1$
$p(T, -1 \,|\, 1, L) = 1$

$p(6, -1 \,|\, 1, L) = 0$
...

No discount

$\gamma = 1$

Random policy

$\pi(U \,|\, 1) = \pi(D \,|\, 1) = \pi(R \,|\, 1) = \pi(L \,|\, 1) = 0$

Bellman Equations: state value function - Recursive value iteration

$$v_\pi(s) = \sum_a \pi(a|s) \sum_{s',r} p(s', r \,|\, s, a) \left[r + \gamma v_\pi(s') \right]$$

k=0

0 T	0 1	0 2	0 3
0 4	0 5	0 6	0 7
0 8	0 9	0 10	0 11
0 12	0 13	0 14	0 T

k=1

0 T	-1 1	2	3
4	5	6	7
8	9	10	11
12	13	14	0 T

$$\begin{aligned} v(1) = {} & 0.25 \cdot 1 \cdot [-1+1 \cdot 0] \; + \\ & 0.25 \cdot 1 \cdot [-1+1 \cdot 0] + \\ & 0.25 \cdot 1 \cdot [-1+1 \cdot 0] + \\ & 0.25 \cdot 1 \cdot [-1+1 \cdot 0] = 4 \cdot 0.25 \cdot -1 = -1 \end{aligned}$$

Deterministic

$p(1, -1 \,|\, 1, U) = 1$
$p(5, -1 \,|\, 1, D) = 1$
$p(2, -1 \,|\, 1, R) = 1$
$p(T, -1 \,|\, 1, L) = 1$

$p(6, -1 \,|\, 1, L) = 0$
…

No discount

$\gamma = 1$

Random policy

$\pi(U \,|\, 1) = \pi(D \,|\, 1) = \pi(R \,|\, 1) = \pi(L \,|\, 1) = 0$

Bellman Equations: state value function - Recursive value iteration

$$v_\pi(s) = \sum_a \pi(a|s) \sum_{s',r} p(s', r \,|\, s, a) \left[r + \gamma v_\pi(s') \right]$$

k=0

0 T	0 1	0 2	0 3
0 4	0 5	0 6	0 7
0 8	0 9	0 10	0 11
0 12	0 13	0 14	0 T

k=1

0 T	-1 1	-1 2	-1 3
-1 4	-1 5	-1 6	1 7
-1 8	-1 9	-1 10	-1 11
-1 12	-1 13	-1 14	0 T

$$\begin{aligned} v(1) = {} & 0.25 \cdot 1 \cdot [-1+1 \cdot 0] \; + \\ & 0.25 \cdot 1 \cdot [-1+1 \cdot 0] + \\ & 0.25 \cdot 1 \cdot [-1+1 \cdot 0] + \\ & 0.25 \cdot 1 \cdot [-1+1 \cdot 0] = 4 \cdot 0.25 \cdot -1 = -1 \end{aligned}$$

Bellman Equations: state value function - Recursive value iteration

$$v_\pi(s) = \sum_a \pi(a|s) \sum_{s',r} p(s', r \,|\, s, a) \Big[r + \gamma v_\pi(s') \Big]$$

k=0

0 T	0 1	0 2	0 3
0 4	0 5	0 6	0 7
0 8	0 9	0 10	0 11
0 12	0 13	0 14	0 T

k=1

0 T	-1 1	-1 2	-1 3
-1 4	-1 5	-1 6	1 7
-1 8	-1 9	-1 10	-1 11
-1 12	-1 13	-1 14	0 T

k=2

0 T	1	2	
4	5	6	
8	9	10	
12	13	14	0

Bellman Equations: state value function - Recursive value iteration

$$v_\pi(s) = \sum_a \pi(a|s) \sum_{s',r} p(s', r \,|\, s, a) \left[r + \gamma v_\pi(s') \right]$$

k=0

0 T	0 1	0 2	0 3
0 4	0 5	0 6	0 7
0 8	0 9	0 10	0 11
0 12	0 13	0 14	0 T

k=1

0 T	-1 1	-1 2	-1 3
-1 4	-1 5	-1 6	1 7
-1 8	-1 9	-1 10	-1 11
-1 12	-1 13	-1 14	0 T

k=2

0 T	-1.75 1	2	
4	5	6	
8	9	10	
12	13	14	0

$$v(1) = 0.25 \cdot 1 \cdot [-1+1 \cdot 0] +$$
$$0.25 \cdot 1 \cdot [-1+1 \cdot -1] +$$
$$0.25 \cdot 1 \cdot [-1+1 \cdot -1] +$$
$$0.25 \cdot 1 \cdot [-1+1 \cdot -1] = 3 \cdot 0.25 \cdot -2+1 \cdot 0.25 \cdot -1 = -1.75$$

Bellman Equations: state value function - Recursive value iteration

$$v_\pi(s) = \sum_a \pi(a|s) \sum_{s',r} p(s', r \,|\, s, a) \left[r + \gamma v_\pi(s') \right]$$

k=0

0 T	0 1	0 2	0 3
0 4	0 5	0 6	0 7
0 8	0 9	0 10	0 11
0 12	0 13	0 14	0 T

k=1

0 T	-1 1	-1 2	-1 3
-1 4	-1 5	-1 6	1 7
-1 8	-1 9	-1 10	-1 11
-1 12	-1 13	-1 14	0 T

k=2

0 T	-1.75 1	2	
4	5	-2 6	
8	9	10	
12	13	14	0

$$v(6) = 0.25 \cdot 1 \cdot [-1+1 \cdot -1] +$$
$$0.25 \cdot 1 \cdot [-1+1 \cdot -1] +$$
$$0.25 \cdot 1 \cdot [-1+1 \cdot -1] +$$
$$0.25 \cdot 1 \cdot [-1+1 \cdot -1] = 4 \cdot 0.25 \cdot -2 = -2$$

Bellman Equations: state value function - Recursive value iteration

$$v_\pi(s) = \sum_a \pi(a|s) \sum_{s',r} p(s', r \,|\, s, a) \Big[r + \gamma v_\pi(s') \Big]$$

k=0

0 T	0 1	0 2	0 3
0 4	0 5	0 6	0 7
0 8	0 9	0 10	0 11
0 12	0 13	0 14	0 T

k=1

0 T	-1 1	-1 2	-1 3
-1 4	-1 5	-1 6	1 7
-1 8	-1 9	-1 10	-1 11
-1 12	-1 13	-1 14	0 T

k=2

0 T	-1.75 1	-2 2	-2
-1.75 4	-2 5	-2 6	-2
-2 8	-2 9	-2 10	-1.7
-2 12	-2 13	-1.75 14	0

$$\begin{aligned} v(6) = {} & 0.25 \cdot 1 \cdot [-1 + 1 \cdot -1] \; + \\ & 0.25 \cdot 1 \cdot [-1 + 1 \cdot -1] \; + \\ & 0.25 \cdot 1 \cdot [-1 + 1 \cdot -1] \; + \\ & 0.25 \cdot 1 \cdot [-1 + 1 \cdot -1] = 4 \cdot 0.25 \cdot -2 = -2 \end{aligned}$$

Bellman Equations: state value function - Recursive value iteration

$$v_\pi(s) = \sum_a \pi(a|s) \sum_{s',r} p(s', r \,|\, s, a) \Big[r + \gamma v_\pi(s') \Big]$$

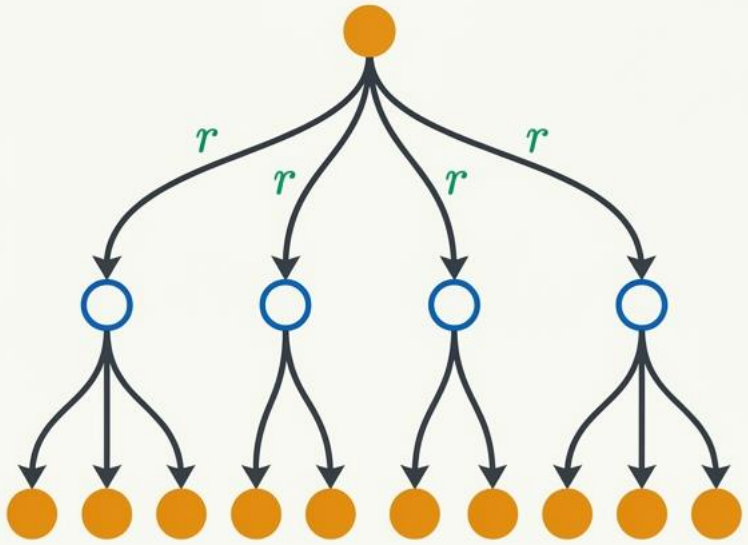
k=10

0 T	-6.1 1	-8.4 2	-9.0 3
-6.1 4	-7.7 5	-8.4 6	-8.4 7
-8.4 8	-8.4 9	-7.7 10	-6.1 11
-9.0 12	-8.4 13	-6.1 14	0 T

Bellman Equations

The Action-Value Bellman Formulation

Backup Diagram for q_π



$$q_\pi(s,a) = \sum_{s',r} p(s',r|s,a)\left[r + \gamma \sum_{a'} \pi(a'|s') q_\pi(s',a')\right]$$

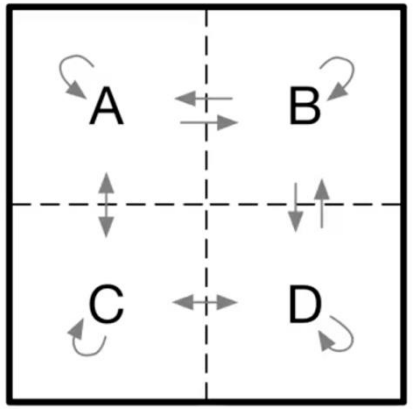
Key Takeaway: Because the action is already chosen at the root, the agent first observes the environment's response, and then averages over its own next possible actions.

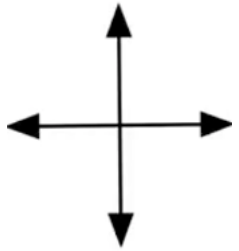
Exercises

Exercise 3.12 Give an equation for v_π in terms of q_π and π.

Exercise 3.13 Give an equation for q_π in terms of v_π and the four-argument p.

Example: Gridworld





$$V_\pi(A) \doteq \mathbb{E}_\pi[G_t \mid S_t = A]$$

$$V_\pi(B) \doteq \mathbb{E}_\pi[G_t \mid S_t = B]$$

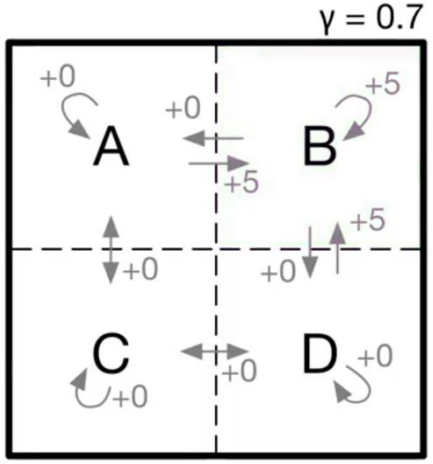
$$V_\pi(C) \doteq \mathbb{E}_\pi[G_t \mid S_t = C]$$

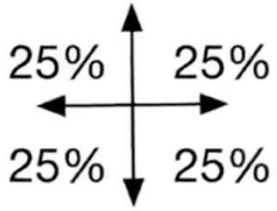
$$V_\pi(D) \doteq \mathbb{E}_\pi[G_t \mid S_t = D]$$

$$V_\pi(s) = \sum_a \pi(a \mid s) \sum_r \sum_{s'} p(s', r \mid s, a)[r + \gamma V_\pi(s')]$$

$$V_\pi(A) = \sum_a \pi(a \mid A)\left(r + 0.7 V_\pi(s')\right)$$

$$V_\pi(A) = \frac{1}{4}(5 + 0.7V_\pi(B)) + \frac{1}{4}0.7V_\pi(C) + \frac{1}{2}0.7V_\pi(A)$$





$$V_\pi(A) = \frac{1}{4}(5 + 0.7V_\pi(B)) + \frac{1}{4}0.7V_\pi(C) + \frac{1}{2}0.7V_\pi(A) \qquad V_\pi(A) = 4.2$$

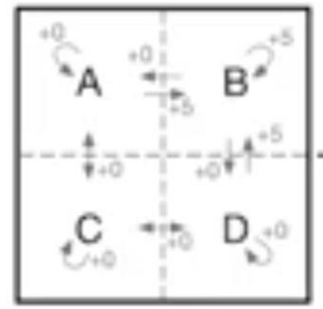
$$V_\pi(B) = \frac{1}{2}(5 + 0.7V_\pi(B)) + \frac{1}{4}0.7V_\pi(A) + \frac{1}{4}0.7V_\pi(D) \qquad V_\pi(B) = 6.1$$

$$V_\pi(C) = \frac{1}{4}0.7V_\pi(A) + \frac{1}{4}0.7V_\pi(D) + \frac{1}{2}0.7V_\pi(C) \qquad V_\pi(C) = 2.2$$

$$V_\pi(D) = \frac{1}{4}(5 + 0.7V_\pi(B)) + \frac{1}{4}0.7V_\pi(C) + \frac{1}{2}0.7V_\pi(D) \qquad V_\pi(D) = 4.2$$

Why Bellman ?

We can only directly solve small MDPs

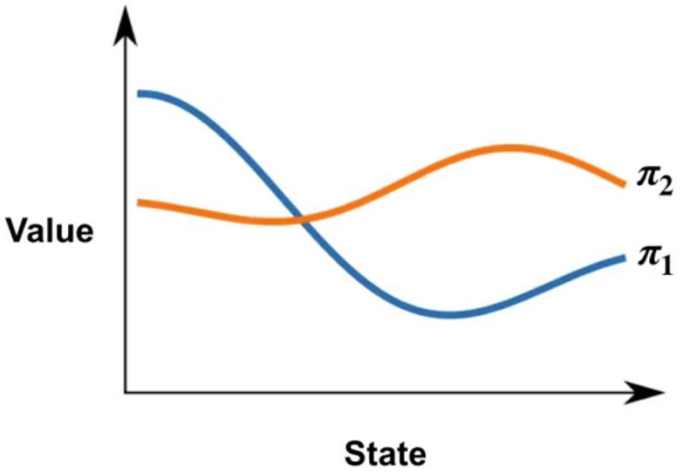


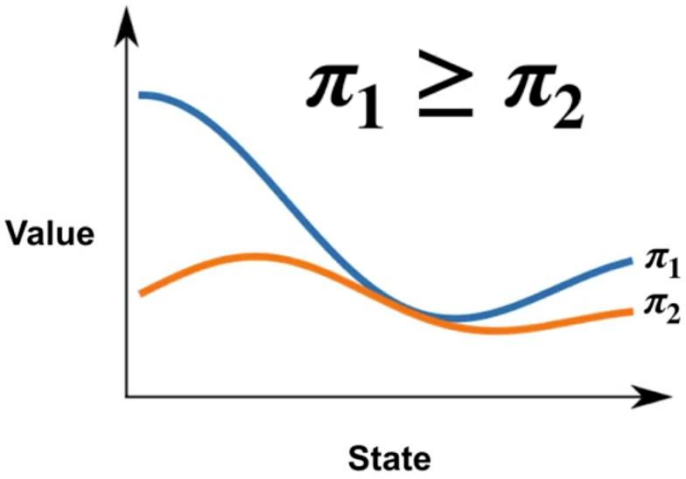
4 Linear Equations → Value Function

10^{45} Linear Equations → Value Function

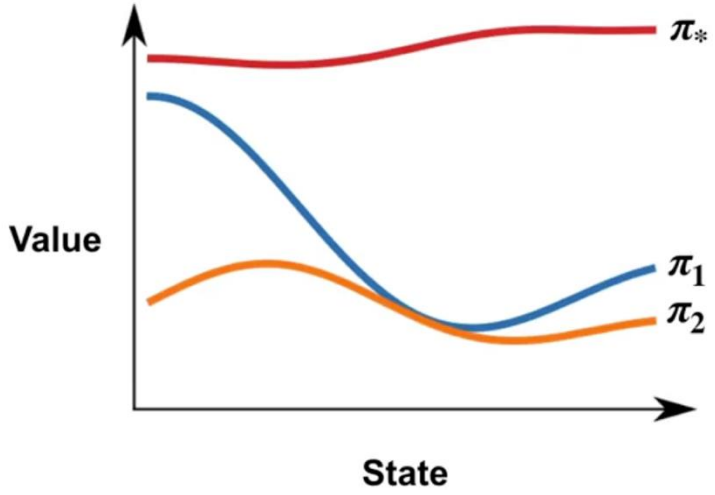
- You can use the Bellman Equations to solve for a value function by writing a system of linear equations
- We can only solve small MDPs directly, but Bellman Equations will factor into the solutions we see later for large MDPs

Optimality: optimal policy

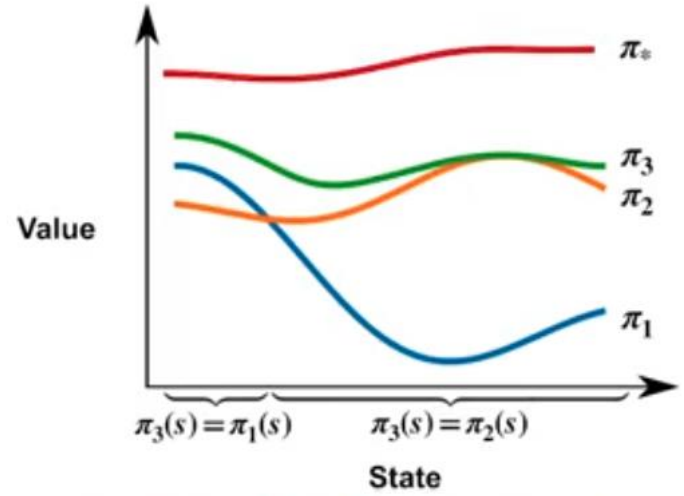




An optimal policy π_* is as good as or better than all the other policies



There is always an optimal policy



There's a rigorous proof showing that there must always exist at least one optimal deterministic policy.

Optimality: Optimal Value Functions

Defining the Ultimate Goal of Optimality

Partial Ordering Constraint:

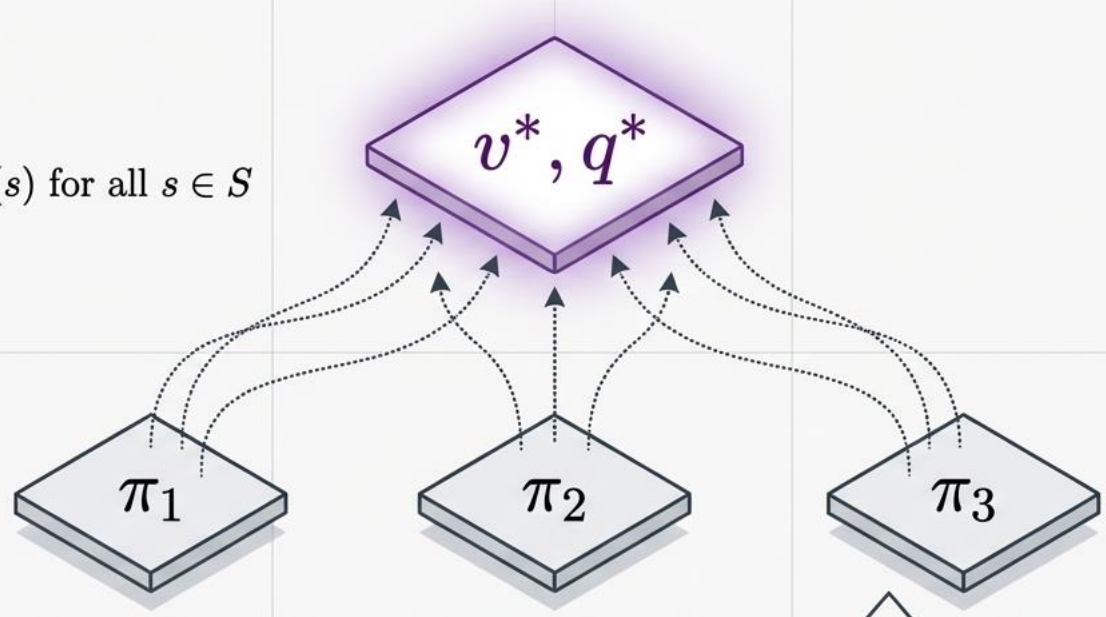
$\pi \geq \pi'$ if and only if $v_\pi(s) \geq v_{\pi'}(s)$ for all $s \in S$

Optimal State-Value:

$$v^*(s) = \max_\pi v_\pi(s)$$

Optimal Action-Value:

$$q^*(s, a) = \max_\pi q_\pi(s, a)$$



Crucial Distinction:

There can be many optimal policies (e.g., multiple equally fast paths through a maze), but they all mathematically **share exactly one optimal value function.**

Optimality: Optimal Value Functions

Recall that

$$v_\pi(s) = \sum_a \pi(a|s) \sum_{s'} \sum_r p(s', r|s, a)\left[r + \gamma v_\pi(s')\right]$$

$$v_*(s) = \sum_a \pi_*(a|s) \sum_{s'} \sum_r p(s', r|s, a)\left[r + \gamma v_*(s')\right]$$

$$v_*(s) = \max_a \sum_{s'} \sum_r p(s', r|s, a)\left[r + \gamma v_*(s')\right]$$

Bellman Optimality Equation for v_*

Recall that

$$q_\pi(s, a) = \sum_{s'} \sum_r p(s', r|s, a)\left[r + \gamma \sum_{a'} \pi(a'|s') q_\pi(s', a')\right]$$

$$q_*(s, a) = \sum_{s'} \sum_r p(s', r|s, a)\left[r + \gamma \sum_{a'} \pi_*(a'|s') q_*(s', a')\right]$$

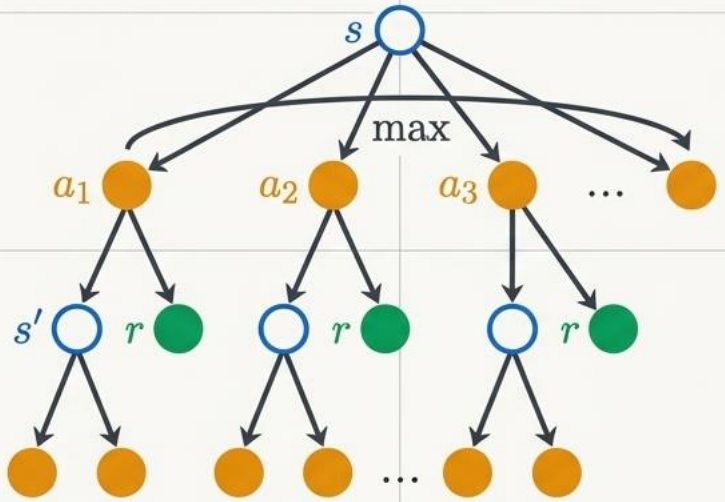
$$q_*(s, a) = \sum_{s'} \sum_r p(s', r|s, a)\left[r + \gamma \max_{a'} q_*(s', a')\right]$$

Bellman Optimality Equation for q_*

Optimality: Optimal Value Functions

The Dynamic Schematic

Solving for Maximum Return: Bellman Optimality



$$v^*(s) = \max_a \sum_{s',r} p(s', r \mid s, a) \left[r + \gamma v^*(s') \right]$$

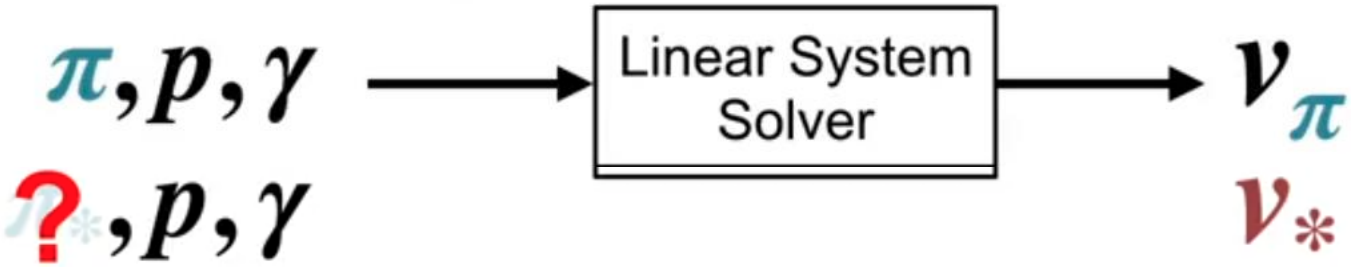
$$q^*(s, a) = \sum_{s',r} p(s', r \mid s, a) \left[r + \gamma \max_{a'} q^*(s', a') \right]$$

Takeaway: An optimal agent does not average its outcomes over bad actions; it exclusively and deterministically picks the highest-yielding action.

NotebookLM

Linear system solver ?

$$v_\pi(s) = \sum_a \pi(a|s) \sum_{s'} \sum_r p(s', r|s, a)\left[r + \gamma v_\pi(s')\right]$$



Not linear

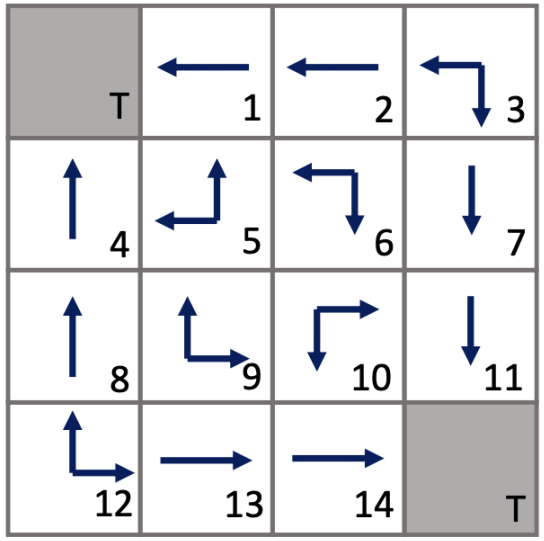
$$v_*(s) = \max_a \sum_{s'} \sum_r p(s', r|s, a)\left[r + \gamma v_*(s')\right]$$

Bellman Equations: state value function - Recursive value iteration

$$v_\pi(s) = \sum_a \pi(a|s) \sum_{s',r} p(s', r \,|\, s, a) \Big[r + \gamma v_\pi(s') \Big]$$

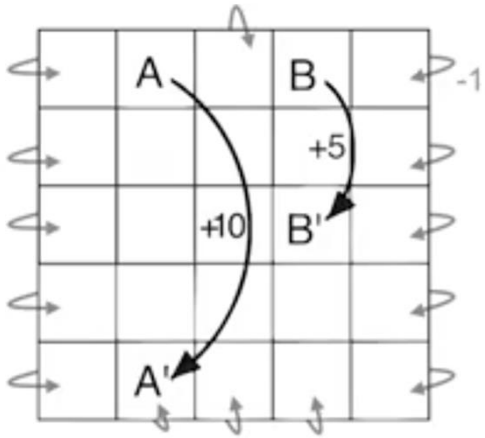
k=10

0 T	-6.1 1	-8.4 2	-9.0 3
-6.1 4	-7.7 5	-8.4 6	-8.4 7
-8.4 8	-8.4 9	-7.7 10	-6.1 11
-9.0 12	-8.4 13	-6.1 14	0 T



Optimal policy π*

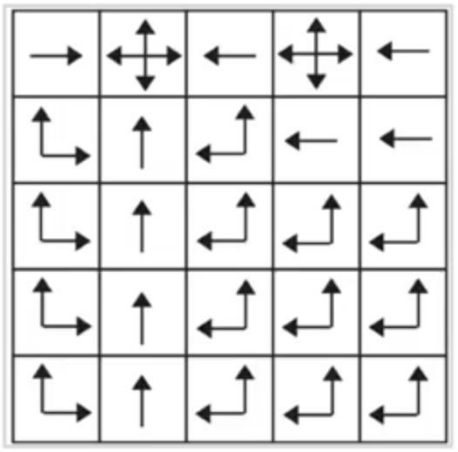
Determining an Optimal Policy : example



$\gamma = 0.9$

22.0	24.4	22.0	19.4	17.5
19.8	22.0	19.8	17.8	16.0
17.8	19.8	17.8	16.0	14.4
16.0	17.8	16.0	14.4	13.0
14.4	16.0	14.4	13.0	11.7

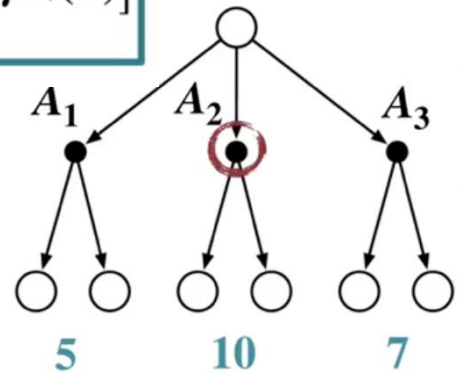
v_*

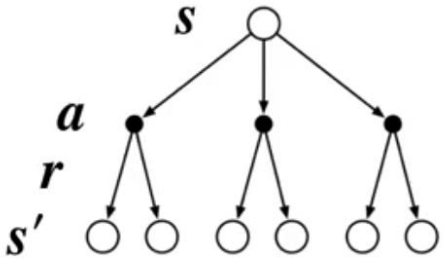


π_*

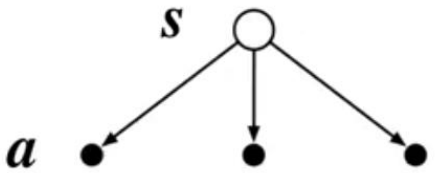
$$v_*(s) = \max_a \sum_{s'} \sum_r p(s', r \mid s, a)\left[r + \gamma v_*(s')\right]$$

$$\pi_*(s) = \operatorname*{argmax}_a \sum_{s'} \sum_r p(s', r \mid s, a)\left[r + \gamma v_*(s')\right]$$





$$\pi_*(s) = \operatorname*{argmax}_a \sum_{s'} \sum_r p(s', r \mid s, a)\left[r + \gamma v_*(s')\right]$$

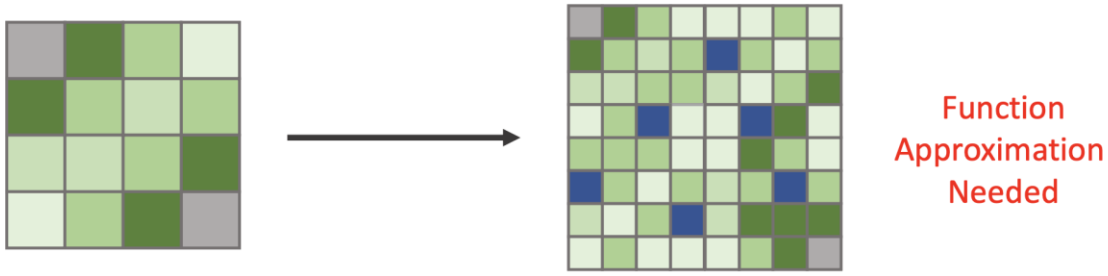


$$\pi_*(s) = \operatorname*{argmax}_a q_*(s, a)$$

Problems

- Optimal policies work well, but in practice hard to achieve
- Works well on small problems
- Computational heavy on large problems
- Memory needy

Curse of dimensionality describes the phenomenon where the feature space becomes increasingly sparse for an increasing number of dimensions



Function Approximation Needed

Determining an Optimal Policy, solving finite MDP

- Remark: In this example, we know exactly how the environment behaves, so we can compute the best action analytically.
- In reinforcement learning, we don't know this — we must learn it from experience.

- **In RL (later):** Model is unknown → we replace:
  - $\sum p\,(\cdot)$ *with samples* (r, s')
- This leads to: TD learning, Q-learning

Dynamic Programming (DP)	Monte-Carlo (MC)	Temporal Difference (TD)
• Well-developed mathematically	• Do not require a model	• Require no model
• Require a complete and accurate model of the environment	• Conceptually simple	• Fully incremental
	• Not well suited for step-by-step incremental computation	• More complex to analyse

References

[1] Sutton, R. S. and Barto, A. G. (2018), Reinforcement Learning: An Introduction (Second edition). MIT Press

[2] Adam White and Martha White, Reinforcement Learning Specialization, University of Alberta, Coursera